
TolTEC Raw-Timestream Conditioning

Learn–Resolve–Apply Filtering, Response,
and Scientific Consequences

Science-Team Rationale

Scientific contract version	v0.1
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Scope of this revision. It defines why RTC operations are selected, how paired x/r readout coordinates are retained, what is learned, what becomes immutable, how level shifts and leakage are represented, how astronomical modes are changed, and which claims remain available afterward. It does not claim implementation conformity, representation fidelity, observational performance, science qualification, validation, or production readiness.

Executive abstract

Raw-timestream conditioning is a scientific method over exact paired raw readout coordinates, not merely a sequence of array operations. Upstream authority maps measured IQ to one x/r pair per detector occurrence. RTC retains both: x is the ordinary science-conditioning path and r is a diagnostic coordinate with independently represented validity. Despiking, level-shift segmentation, atmospheric-template removal, donor replacement, notches, and broad-band filters alter values, response, support, and covariance. Decimation changes temporal identity and can alias rejected modes into the retained band. Each choice can therefore change a point-source peak, effective PSF, centroid, extended-source response, calibration transfer, or scientific eligibility without changing a unit label.

SCI-RTC v0.1/r0.5 organizes those choices as **learn–resolve–apply**. Learn uses a declared population and diagnostics to estimate candidate parameters and uncertainty. Resolve combines that evidence with fixed policy and admits one complete immutable RTC plan. Apply executes exactly that plan and does not relearn or retune. An online adaptive filter is a different, time-varying estimator and remains unavailable unless separately authorized. Missing evidence or tolerances are not passes.

Each resolved response is evaluated against the astronomical temporal spectrum induced by the realized scan velocity, projected beam, source class, cadence, and preceding response. A fixed cutoff is not equally safe for all scan speeds or directions. A notch that removes an interference line also removes whatever astronomical response lies in its width. A detector-static **flxscale** cannot restore removed temporal modes, and calibration transfer across different RTC plans requires explicit response accounting or observational evidence.

The output is an atomic bundle of paired signal, exact mapping and plan identity, response, timing convention, support, influence, validity, uncertainty status, and provenance. RTC coordinates remain raw in every role; raw x is not itself Stokes I and r is neither calibrated nor substituted for x . Absolute **flxscale** and target atmosphere belong to a distinct downstream SCI-CAL product applied only to x . When a required predicate is unresolved, the affected operation or claim is explicitly unavailable. The formal appendices are the sole normative authority; Sections 1–12 explain their scientific meaning.

Reading rule. Narrative text explains the shared authority but cannot weaken a requirement, choose a missing numerical value, or convert an unavailable state into a pass.

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1 Executive summary and present status

RTC is the scientific transformation between an admitted aligned paired x/r detector stream and a conditioned paired stream whose temporal, detector-mixing, and cross-coordinate response is explicit. Its purpose is not to make every sample finite or every spectrum smooth. Its purpose is to address declared contamination and sampling problems while preserving a quantitatively adequate response for the intended astronomical estimator.

The v0.1 scientific method is learned-then-fixed. Evidence may be learned from a declared population, but application begins only after resolve has produced one complete immutable plan. Online adaptation is a different estimator and is unavailable in this revision. Fixed plans use the same resolve and apply discipline even when no parameter is learned from the observation.

Every operation is interpreted through ten questions: purpose, signal model, operator, design parameters, learning, resolution and application, astronomical consequences, calibration consequences, validation, and unavailable states. Those questions are applied below to despiking, donors, masks, notches, broad-band and general filters, state, boundaries, non-finite handling, level shifts, optical leakage diagnostics, raw atmospheric-template removal, and decimation.

The current document is a draft for scientific-owner review. It establishes no production numbers. Open decisions remain open in the ledger; their dependent operations are unavailable. The formal core defines algebraic and semantic authority only. Implementation conformity, representation fidelity, observational performance, science qualification, and production readiness are explicitly unassessed.

Present decision boundary. The owner is being asked to approve the scientific architecture and later resolve the enumerated numerical and policy choices—not to infer those choices from current software or historical practice.

Role-specific RTC plans. The roles share one paired raw signal boundary and one operator order, but they do not inherit one another’s learning population, plan, interval, cadence, or consumer. “Selected” below means owner-resolved for that role; unresolved entries remain unavailable rather than inheriting a software default.

Role	Learning population	Ordered paired RTC operations	Parameters frozen
Beammap	Exact designated Beammap scans	IQ-map parent → despiking/shift segmentation → selected donor/template/filter → phase-zero selection	Complete paired Beammap RTC plan
Pointing	Exact designated Pointing scans	Same paired classes, separately selected; optional classes remain disabled unless selected	Complete paired Pointing RTC plan
OOF	Exact designated OOF scans	Same paired classes, separately selected; optional classes remain disabled unless selected	Complete paired OOF RTC plan
Science	Exact selected science observation or scan set	Same paired classes, separately selected; optional classes remain disabled unless selected	Complete paired Science RTC plan
Diagnostic	Exact named parent and diagnostic sample set	Explicit selected paired subset; no science-role inheritance	Complete diagnostic-only paired plan

Role	Apply interval	Output cadence	Response-sensitive consumer
Beammap	Named half-open Beammap interval	Realized raw RTC grid	SCI-BEAM; later SCI-CAL handoff
Pointing	Named half-open Pointing interval	Realized raw RTC grid	Pointing/SRC and AST consumers
OOF	Named half-open OOF interval	Realized raw RTC grid	OOF/SRC response inference
Science	Named half-open science interval	Realized paired raw RTC grid	SCI-CAL handoff and any explicitly authorized raw-domain consumers; subsequent PTC, VAL, MAP, and FLT use follows their own contracts.
Diagnostic	Exact declared interval	Declared raw diagnostic grid	Named diagnostic consumer only

2 The learn–resolve–apply paradigm

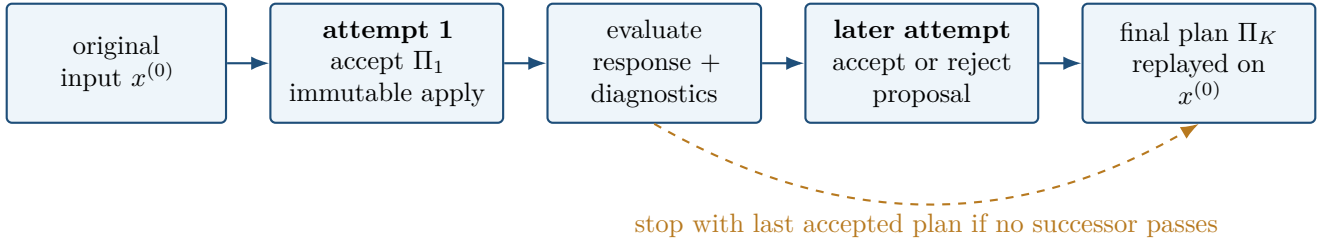
RTC begins with a scientific question: which contaminant or sampling problem must be addressed while retaining adequate astronomical response? The answer is not an operation name. It is a lifecycle that identifies evidence, converts it into a fully specified plan, and applies that plan without hidden adaptation.

Learn. Learn uses a declared population: a separate training interval, a designated part of the same observation, a collection of scans, an observation aggregate, or another named population. It may estimate spike statistics, line centers and widths, filter bands, tap candidates, masks, scan classes, decimation candidates, donor groupings, quality, and uncertainty. For paired streams it may also estimate optical r/x leakage, atmospheric templates, level-shift events, plateau state, and cross-coordinate covariance. Each diagnostic carries estimator, unit, support, selection, and failure state. Learning from the observation later conditioned makes the operator data-dependent. Conditional linear response at fixed state does not eliminate selection bias or parameter-estimation uncertainty.

Resolve. Resolve combines evidence with owner-approved policy. It accepts or rejects every scientific predicate and produces one immutable plan containing sequence, coefficient identity, rates and conventions, order, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation phase, output timing, expected response/support, uncertainty, and downstream compatibility. Missing evidence or tolerance leaves the dependent plan unresolved.

Apply. Apply consumes exactly one resolved plan. It does not move a notch, change a cutoff or tap count, switch donor policy, change decimation, silently fall back, or refit a parameter. Restart restores the same plan and rejects incompatible inputs. Realized state records what ran without rewriting the request or resolved plan.

V0.1 also permits **bounded iterative notch-plan refinement**. This is an outer finite sequence of actual attempts over complete accepted plans, with a separately configured finite maximum. Parameters may change only after one apply has finished and a successor attempt has admitted a new complete plan. A rejected or stop attempt does not advance the accepted-plan index or manufacture an evaluation product. Each apply remains immutable. This is the v0.1 scientific meaning when “adaptive notch” is used; it is not continuous coefficient tracking and is not the online-adaptive estimator of SCI-RTC-DEF-025.



Attempt a learns from an exact population and the evaluation product of the current accepted plan Π_{k_a} . Resolve returns a complete cumulative proposal $\tilde{\Pi}_a$ and a typed accepted, rejected, or stop disposition. Only acceptance advances the plan index and creates a new evaluation product. The initial evaluation is explicitly $x^{\text{eval},(0)} = \mathcal{A}_{\Pi_0}(x^{(0)})$, including when Π_0 is a nonidentity fixed plan. A successor proposal names all retained, modified, removed, or rejected notches and every broad-band, state, mask, edge, timing, and decimation choice. It is never merely “add this notch to whatever ran before.” The final production product normally replays Π_K against the original input, avoiding unintended repeated rounding, padding, state transients, duplicate filtering, stale masks, or ambiguous parentage.

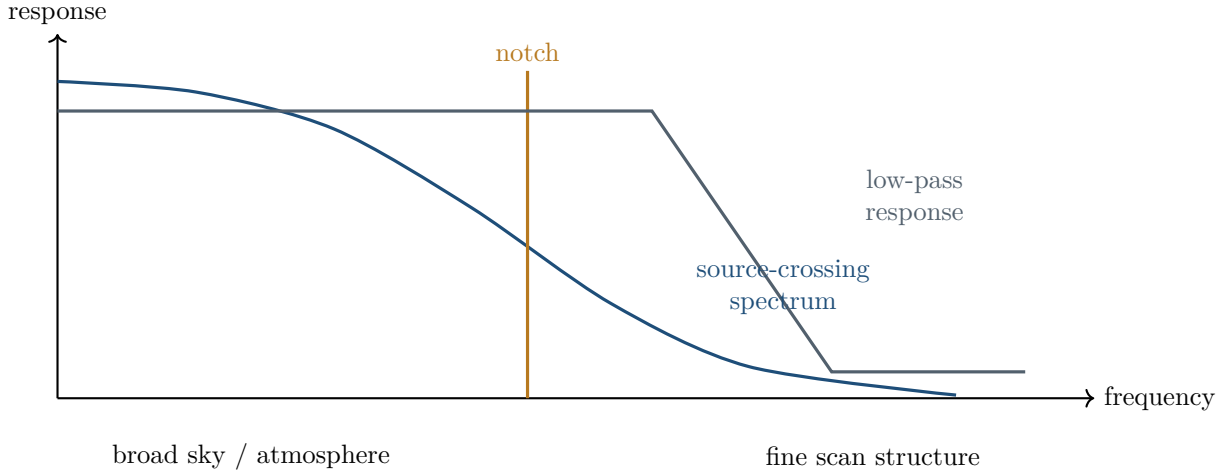
A sequential cascade from the preceding filtered product is a different scientific operator. It is allowed only when selected explicitly and when every immutable intermediate parent, response, state, phase, support, covariance, and uncertainty is retained. Learned-then-fixed and online-adaptive filtering remain different: in the latter, parameters evolve inside one apply pass. V0.1 does not authorize that branch.

Lifecycle test. For each attempt identify its attempt index, current accepted-plan index, learning and parent products, complete proposal, predicates, and disposition. For each accepted plan show that apply did not modify it. Then identify the finite stop cause and the final selected plan. Actual attempts, configured maximum, and accepted plans are different counts; unused maximum attempts are not no-ops. If any link is absent, the refinement claim is unavailable.

3 Temporal frequencies, scan speed, and atmospheric variability

The same temporal frequency represents different angular structure in different scans. For a locally Gaussian effective beam with covariance Σ and angular scan velocity $\mathbf{v}$, the characteristic crossing time is equation (SCI-RTC-EQ-023). It uses the beam projected along the actual scan direction, not merely a scalar array FWHM. Source extent, detector and telescope response, prior RTC stages, and cadence further shape the exact source-times-response spectrum.

High frequencies correspond to finer scan-direction structure. Low frequencies contain broad sky structure together with baselines, atmosphere, and slow instrumental behavior. These do not cleanly separate source and contaminant. A Gaussian is not band-limited; extended emission can share low frequencies with atmosphere; a coherent line can overlap a source crossing. The shorthand $f \sim v/\theta$ is pedagogical only.



Changing speed moves astronomical modes relative to a fixed cutoff or notch; changing direction changes projected beam width. Every filter is assessed over declared speed, direction, beam, source extent, and source class using the exact realized response. Preserved dimensions or units do not establish preserved peak, PSF, centroid, extended response, or absolute mode.

Atmospheric fluctuations are also structured in time, detector, and readout coordinate. A raw atmospheric template may be scientifically useful because it can expose common optical response in x and r , but it is not the sample-dependent target-atmosphere correction owned by SCI-CAL. Its learning population, estimator, frequency support, detector grouping, masks, quality, uncertainty, and applicability all belong to the RTC plan.

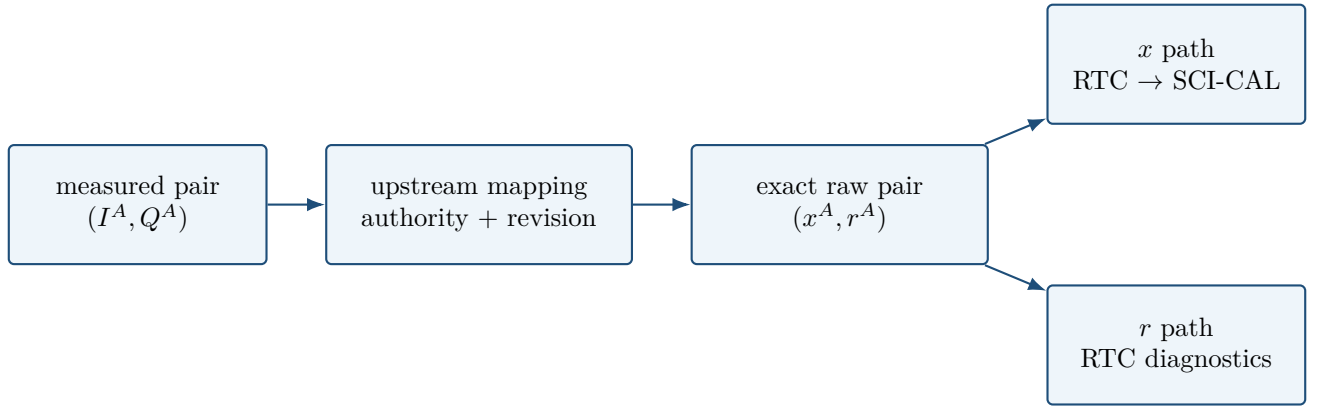
Template removal and temporal filtering generally do not commute. The owner has selected template removal before later RTC filters. The response of that composition, or a quantitative bound over its admitted domain, is required; reversing the order is not an equivalent implementation detail. Low-frequency sky modes may overlap the learned atmospheric subspace, so the validation domain must include compact and extended sources across scan speeds and directions. Missing response authority makes the affected transfer claim unavailable rather than “approximately preserved.”

4 Paired IQ-to- x/r readout coordinates

The acquired scientific object is an ordered pair, not two streams that may later be joined by row number. Upstream mapping authority converts measured in-phase and quadrature coordinates into x_{dj}^A and r_{dj}^A with one detector, sample, stream, and mapping-revision identity. RTC consumes that mapping and its coefficient, offset, applicability, validity, and provenance; it does not infer, repair, or silently replace it.

Both members are required for every admitted occurrence. A missing partner is malformed input, but numerical or scientific validity is represented independently: valid r does not repair invalid x , and invalid r does not erase an otherwise valid x . Exact pair identity survives segmentation, filtering, selection, serialization, and restart. The notation m_{native} denotes a native table row so that the physical r coordinate is never confused with a row index.

The roles of the members are intentionally asymmetric. Raw x is the ordinary science-conditioning path, but it is not itself Stokes I. Raw r is a diagnostic coordinate that can reveal optical response, level shifts, electronics behavior, and shared selection. V0.1 does not authorize automatic r -to- x correction, calibration of r , use of r as an x donor, or substitution of r for invalid x . Polarimetry remains outside scope.



Paired statistics retain the full x/r covariance, common support, and any shared selector. A block-diagonal or independent model requires evidence; it is not implied by coordinate names. This paired boundary applies equally to Beammap, Pointing, OOF, Science, and diagnostic-only roles.

5 Despiking, level shifts, and donor replacement

Despikes and level shifts are different contaminants. An isolated spike is a localized departure from the intended signal model. A level shift is a persistent paired offset change with a transition, pre/post plateaus, and a new coherent processing segment. Treating a shift as a cluster of spikes can replace real data and leave a biased plateau; treating a spike as a shift can unnecessarily reset a long filter.

The owner-selected early suborder is explicit: detect spikes and apply the admitted sample-replacement policy first, then estimate level shifts on that repaired paired stream. This choice prevents isolated extremes from driving the shift estimator. The resolved plan still records the despike statistic, threshold, support, mask, false-positive control, donor topology, and exact transition masks used by the later estimator. An alternative suborder is not implied by this contract.

A level-shift event records detector, transition time or interval, separate x and r amplitudes or typed unavailability, estimator, support, quality, uncertainty, and cause. Detection does not itself authorize correction. Each admitted event creates a transition mask and owner-resolved pre/post guards, delimits coherent segments, and invokes an explicit reset or carry rule. Multiple events retain ordered identities, even when guards overlap or the intervening segment is too short; a selected conflict and minimum-segment policy determines the disposition.

Plateau estimates can diagnose pre/post state and help assess the event, but remain inert or advisory unless selected as policy input. V0.1 does not authorize plateau-based cross-segment offset stitching. Such stitching would be a new response-changing operator with its own support, covariance, provenance, and validation burden.

A **donor** is an admitted detector-time measurement used to construct a continuity surrogate for a rejected target x cell. The target retains its identity and replacement cause while donor information and covariance remain traceable. Under the approved raw convention, compatible valid **flxscale** values permit the factor ϕ_q/ϕ_d ; this transfers convention, not sky identity or calibration. The paired r member is never an x donor. Beammap donor factors come from identified prior authority, never the new factor derived from the same Beammap lineage.

The exact representative occurrence for a phase-zero output is (d, Mn) in the pre-replacement acquired stream. If that occurrence is synthesized or replaced, the output is universally excluded as an independent detector measurement. A donor, maximum response coefficient, or support centroid does not replace this definition. Nonrepresentative influences preserve their causes and are routed to each downstream consumer's own eligibility policy.

6 Optical leakage diagnostics from atmosphere and bright sources

The r coordinate is not presumed optically dark. A source-dependent local model permits both x and r to respond, and equation (SCI-RTC-EQ-032) defines a diagnostic ratio only where the x response is valid and nonzero. That ratio is not a universal detector correction. It is an evidence product with a source class, estimator, support, selection, uncertainty, validity, and response convention.

Atmospheric variability provides one high-occupancy optical diagnostic. Its estimator must distinguish the selected atmospheric template or population, temporal and frequency support, detector grouping, masks, common-mode model, quality, and uncertainty. Because the atmosphere can share modes with sky signal and electronics, coherence alone does not establish optical origin.

A bright compact source provides a second diagnostic with different spatial and temporal support. It binds the source model, scan geometry, projected beam, background treatment, masks, fit, uncertainty, and source-class applicability. An atmosphere-derived and bright-source-derived value keep separate parentage. Agreement is useful evidence; disagreement is also a result. They are not averaged, substituted, or promoted as mutually validating without an explicit comparison model.

Each diagnostic states whether it is scalar, band-limited, or frequency dependent. A scalar fitted over one band cannot silently become authority outside it. If atmospheric-template removal or subsequent filtering changes the two coordinate responses differently, pre- and post-operation leakage estimates are distinct products. Their comparison includes the composed response and common support.

No leakage diagnostic in v0.1 authorizes automatic r -to- x correction, calibration of r , or a new donor route. Its immediate scientific purpose is to expose readout-coordinate response, detect regime changes, and provide traceable evidence for a later owner decision.

7 Notch and broad-band filtering

A notch suppresses a narrowband coherent disturbance while removing the same temporal modes from any astronomical signal. A resolved notch declares center, frequency/rate convention, width or Q, depth, family/order, poles/zeros where applicable, phase, group delay, state, transient, guard, applicability, uncertainty, and drift tolerance. A disappearing line is not by itself evidence that astronomical transfer, centroid, PSF, covariance, or edge support remains adequate.

Bounded iterative refinement is an outer succession of complete immutable plans. Each actual attempt learns from the evaluation of the current accepted plan, proposes a complete cumulative successor, and records an accepted, rejected, or stop disposition. Acceptance alone advances the plan index. The final plan normally replays once against the original input. A configured maximum is only a bound: an early stop creates no fabricated no-op attempts. Maximum-attempt, oscillating, or parameter-unstable termination is not convergence.

A low-pass may reduce high-frequency noise or prepare for decimation but can attenuate fast source crossings, broaden the PSF, and increase covariance. A high-pass can suppress drift or broad atmosphere while removing extended sky, beam wings, baselines, and absolute modes and producing negative lobes. A band-pass inherits both boundaries. The combined exact response, not the class label or preserved unit, defines what remains.

Learn identifies the contaminant and candidate design over a declared population. Resolve freezes every band, ripple, attenuation, family, order, precision, phase, state, mask, edge, and applicability field. Apply does not retune. Validation varies interference parameters, speed, projected beam, source class and extent, measuring residual, amplitude, centroid, PSF, extended transfer, ringing, support, covariance, and usable fraction.

8 Filter order, phase, registration, and composition

FIR tap count is a scientific design result, not an arbitrary integer. Equation (SCI-RTC-EQ-025) chooses the smallest admitted order that passes transfer, alias, phase, support, guard, precision, and realizability predicates. An approximate family formula can initialize a search but cannot replace exact coefficients and response.

More taps can sharpen a transition while increasing support, covariance footprint, edge loss, ringing, and the range affected by a replacement or level-shift guard. Quantization can change ripple, attenuation, notch placement, and stability. Forward/backward filtering changes magnitude, effective order, state, and edge behavior even when ideal interior phase is zero.

Delay creates scan-direction displacement through equation (SCI-RTC-EQ-027). A phase-zero selected grid point does not imply that the preceding response is centered there. Every role resolves zero phase, shifted time, coordinate correction, response-aware downstream use, or a quantitative bound on residual centroid and PSF effect.

Composition order is equally scientific. Despise replacement precedes level-shift estimation; admitted level shifts define state boundaries; selected raw atmospheric-template removal precedes later temporal filters; and all RTC operations precede SCI-CAL. Each response is composed in that order. Commutation may be claimed only over a declared domain with an exact identity or owner-resolved bound.

Validation varies order/taps, family, precision, bands, shift boundaries, template strength, scan speed, beam, and source class while measuring amplitude, shape, delay, ringing, covariance, registration, and usable support. Coefficient algebra alone is not science qualification.

9 Decimation and alias control

V0.1 phase-zero downsampling is point selection $v^{\text{preD}}[Mn]$, not averaging and not anti-alias filtering. It changes cadence, Nyquist frequency, temporal identity, support, covariance, aliasing, and coordinate registration for both members of the exact pair. The preceding composed filter determines the rejected-band power that remains to fold.

Equation (SCI-RTC-EQ-014) includes every folded image. Before resolving M , the exact prefilter and any preceding atmospheric-template operation must meet approved transfer and alias bounds over the declared source domain. A Gaussian is not band-limited, so safe means bounded residual transfer, not perfect preservation. Factor choice uses cadence, smallest projected crossing time, maximum valid scan speed, source class, output requirements, timing accuracy, and downstream compatibility.

The representative source occurrence is exactly (d, Mn) in the acquired pre-replacement stream. That identity is distinct from the assigned output time, the full transitive support, and the largest response coefficient. It drives the direct synthesized/replaced exclusion without converting every nonrepresentative influenced sample into universal ineligibility.

Selected time t_{Mn} does not erase earlier delay or broad support. Tests include paired impulses and tones, on/off-bin frequencies, varied scan speeds/directions, smallest-beam sources, incomplete strides, level-shift boundaries, reset behavior, and exact alias response.

10 Masks, segment boundaries, state, and non-finite handling

A mask is operator control, not acquisition validity or confidence. Source, spike, level-transition, guard, coordinate, and learning masks retain distinct causes, geometry, support, dilation, and boundary conventions. Invalid or wrong-frame coordinates do not become outside-mask samples. A flag does not automatically become a mask, zero weight, or universal eligibility decision.

Gaps, resets, and selected level shifts delimit coherent segments. FIR extension, rejection, reflection, zero padding, and local renormalization are different operators. IIR state requires explicit initialization, reset or carry, transient, stability, and terminal state. State shall not silently cross a declared shift boundary. A guard removes or types outputs but does not prove a residual tail negligible without a quantitative criterion.

The realized event and segment state includes ordered shift identities, transition intervals, paired amplitudes, masks, guards, plateau diagnostics, short-segment disposition, and reset/carry decisions. Optional operation classes remain disabled unless the role plan explicitly selects them.

A non-finite sample, coordinate, mapping value, factor, coefficient, or state is rejected, represented unavailable over its exact influence, or handled by an authorized prior recovery rule. It never silently becomes zero. Tests place impulses, sources, spikes, shifts, masks, and non-finites at every boundary and compare split/rejoined execution against the declared state semantics.

11 Calibration handoff and downstream interpretation

RTC distinguishes detector-static **flxscale**, an optional raw RTC atmospheric template, sample-dependent target-atmosphere correction, detector mixing, and temporal filtering. Their names do not make them commute. RTC remains raw through the complete paired plan. A compatible valid **flxscale** ratio may transfer donor convention within raw x replacement, but it is not absolute calibration and never involves r .

Only after the complete RTC bundle exists may SCI-CAL consume the exact conditioned x member and apply absolute **flxscale** and target atmosphere. The r member remains in the RTC bundle for traceability and diagnostics but receives neither operation. This boundary does not assert that raw x is Stokes I.

Filtering can preserve dimensions while changing source peak, PSF, extended response, phase, centroid, and covariance. Calibration cannot restore removed modes. A Beammap-derived factor belongs to its complete RTC lineage; a target conditioned with a different plan needs explicit response calculation, renormalization, or observational evidence before calibration transfer is scientifically available.

The Science consumer route is: “SCI-CAL handoff and any explicitly authorized raw-domain consumers; subsequent PTC, VAL, MAP, and FLT use follows their own contracts.” Pointing, OOF, Beammap, and diagnostic consumers also bind their exact paired parent and role plan. RTC does not assign their eligibility, weighting, mapping, or source-estimation policies.

Beammap circularity guard. New **flxscale** derived from a Beammap cannot be used as donor scaling inside that same Beammap RTC lineage; donor factors come from explicit pre-existing authority.

12 Covariance, support, eligibility, validation, and open decisions

Output dependence expands through paired mapping, alignment, despiking selection, donors, level-shift segments, atmospheric templates, taps or state, boundaries, and sampling. Support—not storage shape—governs information flow. Reused donors correlate targets, overlapping filters correlate neighbors, shared x/r selectors create cross-coordinate covariance, and an IIR links every output to its declared state boundary.

At fixed realized state, the conditional moment equations propagate an admitted joint x/r model through the same operator. Data-dependent selection, leakage estimation, shift detection, learned templates, or notch parameters add selection and model variation. Conditioning does not make those terms zero. Unknown covariance remains unavailable rather than diagonal, white, independent, or zero.

Eligibility differs from computability. V0.1 universally excludes an output whose exact representative occurrence (d, Mn) was synthesized or replaced as an independent detector measurement. Elsewhere in full support, RTC preserves value, response, typed cause, linkage, and influence while each consumer applies its own declared policy. Finite paired TOD without exact plan, support, response or typed unavailability, influence, validity, uncertainty status, and provenance is incomplete.

Algebraic fixtures are necessary but not sufficient. Validation varies the scientific design choices and measures the observables they can change.

Study	Variations	Measurements
Notch	center error, drift, width, depth, Q/order, state, direction, overlap	residual line, sky transfer, ringing, transient, amplitude, centroid, covariance, edge loss
FIR/bands	taps, edges, ripple, attenuation, family, precision, speed, beam, source extent	peak/shape, extended transfer, delay, ringing, alias, covariance, usable fraction
Donor	spikes, invalids, factors, sky mismatch, noise, reuse, sources, mask edges	stabilization, bias, covariance, influence extent, scientific data loss
Decimation	on/off-bin tones, speeds/directions, smallest beam, delay, aliases, strides, resets	retained/aliased transfer, timing, coordinate shift, PSF, support, edges
Paired/leakage	mappings, independent validity, atmosphere/source class, scalar/frequency response	pair fidelity, x/r covariance, leakage bias, pre/post response, unavailable states
Level shift	spike adjacency, amplitudes, multiple events, guards, short segments, reset/carry	event accuracy, false detection, plateau bias, state leakage, response and data loss

Each study preregisters domain, estimator, tolerance and falsifier. Passing coefficient algebra does not establish science qualification. Observational performance, science impact and production readiness are separate layers.

The owner ledger retains unresolved numerical or methodological authority for mapping tolerances; independent validity; leakage estimators and frequency status; level-shift thresholds, guards, amplitudes, multiple-event handling, and plateau diagnostics; atmospheric-template estimation and noncommutation bounds; learning populations and freeze points; filter designs; edges/state; decimation/alias; registration; uncertainty; and bundle representation. The paired stream, no automatic r correction, despiking-before-shift order, atmosphere-before-filter order, x -only CAL handoff, representative occurrence, and downstream ownership are already resolved architecture and are not reopened as numerical questions. Online adaptation remains deferred. An open entry blocks only its named operation or claim and never authorizes a default or silent fallback. Cross-package follow-up routes CAL, BEAM, AST, PTC, VAL, MAP and filtering questions without letting RTC assume their answers.

Success criterion. For every operation, a scientist can explain paired input authority, purpose, learned evidence, resolved numerical choice, immutable application, level-shift/segment state, leakage and atmosphere treatment, changed astronomical modes, response and calibration effects, validation evidence, and remaining unavailable claims.

A Shared normative authority and notation

Normative core stamp: SCI-RTC-v0.1/r0.5. The six files in `src/common/` form one authority. Text outside this shared core is explanatory or conformance guidance and cannot alter a definition, equation, assumption, requirement, prediction, unavailable state, or owner decision.

SCI-RTC begins with admitted aligned paired raw detector-coordinate streams **xs** and **rs** and ends with an atomic, role-specific raw conditioning bundle. The x coordinate supplies the ordinary total-intensity analysis path; the paired r coordinate supplies readout, leakage, artifact, and Tune-health information. Raw x is not itself a Stokes parameter. ALIGN, the Tune/readout mapping authority, AST, and SCI-BEAM retain authority for the upstream meanings they produce; SCI-CAL may calibrate only the conditioned x path of an exact RTC bundle. PTC, VAL, MAP, BEAM, NOI, MAP-003, FLT, SRC/MODE, and FRUIT retain authority for their downstream science. This contract owns the realized conditioning order and its scientific consequences; it does not infer current software behavior.

The words **shall** and **shall not** identify requirements. **Required unavailable** means that the affected product or claim shall not be reported as complete. **Optional unavailable** is a typed, scientifically inert state. Missing, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted remain distinct states.

A.1 Indices, sets, and role labels

Symbol	Normative meaning
o	Immutable observation identity.
$d, q \in \mathcal{D}_o$	Exact admitted detector occurrences; d is a target and q may be a donor. A detector index or row number alone is not this identity.
m_{native}	Native acquisition-row index; it is not a physical r coordinate or an aligned or conditioned slot.
$j \in \{0, \dots, N-1\}$	Ordered aligned input slot on the admitted ALIGN grid.
$n \in \{0, \dots, N_y-1\}$	Ordered phase-zero RTC output slot.
$\mathcal{R}$	RTC product role, including Pointing, OOF, Science, Beammap, diagnostic-only, or another separately approved role; role changes policy and consumers, not the raw signal identity.
x_{dj}^A, r_{dj}^A	Exactly paired admitted aligned raw KID readout coordinates. x supplies the ordinary total-intensity analysis path; r supplies readout/leakage/artifact/Tune-health evidence. They share occurrence identity, cadence, phase, and mapping lineage but retain independent value validity.
$\mathbf{v}_{dj}^{IQ}$	Upstream orthogonal microwave transmission components $(I_{dj}, Q_{dj})^\top$.
$\mathbf{u}_{dj}^{xr}$	Paired raw readout vector $(x_{dj}, r_{dj})^\top$ produced by the admitted Tune-dependent mapping.
ζ	Complete Tune and $I, Q \rightarrow x, r$ mapping identity, reference, normalization, validity domain, epoch, and uncertainty state.
χ_{dj}	Detector-static flxscale comparison coordinate used only to define compatible donor transfer; it is not a calibrated RTC product.
u_{dj}	Raw $\Delta f/f$ value immediately before the ordered RTC filter chain.
v_{dj}^{preD}	Final raw $\Delta f/f$ stream after every admitted RTC replacement/filter stage and immediately before phase-zero point selection.
y_{dn}^{RTC}	Conditioned raw $\Delta f/f$ RTC output at the selected phase-zero slot.
c_{dn}^{CAL}	Distinct downstream SCI-CAL output, when authorized: top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam.
ϕ_d	flxscale for one exact detector occurrence under the declared multiplicative convention and calibration domain.
A_α	Imported, realized ALIGN mapping with state α ; its scientific meaning is not owned by RTC.
C_κ^{CAL}	Downstream SCI-CAL operator with state κ , including selected absolute flxscale and target-atmosphere correction; RTC does not apply it.
B_ω	Realized despike/replacement operator conditioned on selection state ω .
F_{k, θ_k}	Realized ordered FIR, IIR, notch, or admitted mask-controlled stage k .
$D_{M,0}$	Integer phase-zero point selector of factor $M \geq 1$.

Symbol	Normative meaning (continued)
Ω	Complete realized conditioning state, including all upstream bindings and every selected stage, exception, state, edge rule, and output interval.
$\mathcal{L}$	Declared learning population with exact identity, selection, support, masks, exclusions, and compatibility domain.
Λ	Learned evidence and uncertainty produced by declared diagnostics; it is not authorization to apply.
Π	One immutable resolved RTC plan admitted from learned evidence and fixed scientific policy.
$\mathcal{A}_\Pi$	Apply operator that consumes plan Π without changing it.
$a \in \{1, \dots, A\}$	Refinement-attempt index; A is the number of attempts actually executed.
$A_{\max}$	Finite owner-approved maximum attempt count, with $A \leq A_{\max} < \infty$.
$k \in \{0, \dots, K\}$	Accepted-plan index; $K \leq A$ is the final accepted-plan index.
k_a	Accepted-plan index available at the start of attempt a .
$\tilde{\Pi}_a$	Complete cumulative successor plan proposed by attempt a ; it is not an accepted plan unless its disposition is accepted.
δ_a	Typed attempt disposition in $\{\text{accepted}, \text{rejected}, \text{stop}\}$.
Π_k	Complete cumulative accepted plan k ; Π_0 is the admitted initial fixed or identity plan.
$x^{\text{eval},(k)}$	Exploratory evaluation product for accepted plan k , normally obtained by applying Π_k to original admitted input $x^{(0)}$.
ρ_{dn}	Representative aligned occurrence (d, Mn) on the raw paired ALIGN parent grid, before replacement and temporal propagation.
$\alpha_{x,d}, \alpha_{r,d}$	Local optical-response coefficients in the paired readout coordinates over a declared signal and support domain.
ϵ_d, ψ_d	Optional scalar optical-leakage ratio $\alpha_{r,d}/\alpha_{x,d}$ or response-direction angle; neither parameterization is universal.
$a_g(t)$	Declared atmospheric optical-mode template or latent common mode for network g .
$\tau_g, \Delta \mathbf{b}_{d,g}$	Network level-shift change time and detector-specific paired amplitude $(\Delta x_{d,g}, \Delta r_{d,g})^\top$.
$I_h, \mathcal{P}_{d,h}$	Coherent plateau interval h and its role-specific admissibility state.
$K_{dn,qj}$	Local detector-time response of one output cell to one acquired input cell.
$\mathcal{S}_{dn}$	Exact transitive input support of output (d, n) , including alignment, donors, filters, state, and selection.
$\mathcal{I}_{dn}$	Cause-preserving transitive influence set for output (d, n) .

A.2 Identity and interval conventions

Observation, Tune, array, network/interface, detector occurrence, selected APT row/binding, native row, aligned slot, scan, science interval, context interval, conditioned output slot, stage, and product identities are non-interchangeable. Array identity is one of **a1100**, **a1400**, and **a2000**; array ID, array index, network ID, detector index, and container position do not substitute for one another. Cross-artifact identity requires an explicit occurrence- and product-scoped binding. Equal row numbers do not establish identity.

Primary detector timestreams are samples on rows and detectors on columns unless a product explicitly declares another ordered shape. Every interval is half-open. Science windows, physical scans, processing chunks, filter context, output selections, and learned-plan scan sets retain separate identities. Native, aligned, and output indices never silently renumber one another.

Time is seconds on the admitted ALIGN-assigned grid. The approximately 8-ms rate relationships and the $0.5\times$, $2\times$, and $4\times$ family are conditional software-grid facts, not proof of a detector integration event, absolute epoch, or astrometric correction. Coordinate-dependent Pointing, OOF, and Beammap controls use a declared AltAz tangent plane; Science controls use a declared equatorial J2000 TAN relation. RTC performs no implicit frame conversion.

A.3 Logical and availability notation

$\mathbb{I}[\cdot]$ denotes an indicator, $\mathbb{E}$ expectation, Cov covariance, and $\text{supp}(\cdot)$ operator support. The logical symbols $\wedge$, $\vee$, and $\neg$ have their usual meanings. A quantity decorated available is present with its required identity, domain, and validity; a quantity decorated unavailable is explicitly unavailable. Neither zero nor NaN is an availability encoding.

End of notation; shared definitions follow without a change of authority.

B Normative definitions

SCI-RTC-DEF-001 – Admitted aligned paired stream.

An immutable observation-scoped pair of **xs** and **rs** arrays. Every admitted x has exactly one r partner with the same observation, Tune, array, network/interface, detector occurrence, native acquisition occurrence, aligned slot, cadence, phase, and mapping lineage. Missing a partner is malformed required input. The paired samples retain independent finiteness, origin/synthesis, support, validity, and uncertainty states.

SCI-RTC-DEF-002 – RTC product-role contract.

The tuple $(\mathcal{R}, s_{\mathcal{R}}, b_{\mathcal{R}}, \mathcal{V}_{\mathcal{R}}, \mathcal{C}_{\mathcal{R}})$ naming role, sign, reference/baseline, valid domain, and required consumers. Every RTC role consumes paired raw **xs/rs**; only conditioned x is eligible for a later SCI-CAL handoff. A calibrated downstream detector signal is a distinct SCI-CAL product c^{CAL} in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention; it is not another meaning of **xs**.

SCI-RTC-DEF-003 – Exact detector occurrence.

An observation-local layered binding containing observation, Tune, network/interface, network-local tone or admitted row slot, selected APT identity/digest, and row association. A transient column number is only a locator.

SCI-RTC-DEF-004 – Downstream CAL handoff.

An immutable RTC-to-SCI-CAL boundary carrying the raw conditioned parent, detector binding, complete RTC plan/response/support, selected **flxscale** authority, target-atmosphere lineage, validity, uncertainty availability, and provenance required by SCI-CAL. Absolute **flxscale** and target atmosphere are applied only after RTC; RTC does not derive, apply, repair, or promote the calibrated product. Because target-atmosphere correction is sample dependent and follows temporal filtering, a handoff claim also requires complete response accounting over RTC support or a preregistered owner-approved noncommutation bound; output-time correction is not an automatic inverse of a filtered atmosphere operator.

SCI-RTC-DEF-005 – Compatible flxscale pair.

Two finite factors ϕ_q and ϕ_d that are valid for the exact donor and target detector occurrences under the same multiplicative calibration convention, quantity domain, reference plane, and applicable support, with $\phi_d \neq 0$. Compatibility is not established by matching row numbers or finite values alone.

SCI-RTC-DEF-006 – Selected conditioning policy.

A versioned, admitted specification of despiking detection, donor eligibility, donor scoring and ties, replacement and fallback, source protection, filter stages and order, exact coefficients and normalization, numeric precision, state/reset, boundaries/edges, non-finite handling, sampling mode, phase-zero factor, flags, influence aggregation, diagnostics, and failure behavior. It is selected authority, not an implementation default.

SCI-RTC-DEF-007 – One-way state record.

The ordered record $(\Theta^{\text{req}}, \Theta^{\text{eff}}, \Theta^{\text{obs}}, \Theta^{\text{learned}}, \Theta^{\text{resolved}}, \Theta^{\text{real}})$. Requested state preserves the accepted request; effective state is the admitted typed plan; observation-resolved state binds exact observation facts; learned state records evidence when learning occurs; resolved state is the immutable admitted plan for every fixed or learned operation; and realized state records what ran. A later state never rewrites an earlier state.

SCI-RTC-DEF-008 – Realized conditioning operator.

The chronological composition of every admitted paired-channel admission, alignment, raw-domain replacement, level-shift transition mask and plateau segmentation, mask, FIR/IIR/notch, boundary, reset state, non-finite, and phase-zero sampling action, including all affine state terms and exception intervals. Annotations that affect values, support, influence, or availability are part of the operator.

SCI-RTC-DEF-009 – Phase-zero sampling.

Point selection of the pre-sampling x stream at input indices $j = Mn$ for one integer factor $M \geq 1$. It is not an

arithmetic mean, block aggregate, or implicit anti-alias filter. Its representative occurrence is exactly (d, Mn) on the raw paired aligned parent before replacement and filtering; that identity is not a largest-coefficient claim.

SCI-RTC-DEF-010 – Complete realized response.

An exact local or factorized representation that includes every enabled response-changing stage, detector mixing, mask, state, edge rule, exception, and sampling phase on its declared support. If such a representation is not truthfully available, response state is unavailable; a partial kernel is not complete response.

SCI-RTC-DEF-011 – Representative source occurrence.

For output (d, n) , the raw aligned paired occurrence (d, Mn) before replacement and temporal propagation. It may be original, ALIGN-synthesized, or subsequently RTC-replaced. This identity, rather than the largest realized coefficient, governs independent-exposure exclusion.

SCI-RTC-DEF-012 – Transitive influence.

The cause-preserving closure from every original, synthesized, replaced, masked, edge, state, and non-finite source cell through all nonzero realized alignment, donor, filter, state, and sampling dependencies to an output. Influence is broader than the selected point and may cross detector columns.

SCI-RTC-DEF-013 – Direct exclusion and consumer eligibility.

An output whose representative occurrence of SCI-RTC-DEF-011 is ALIGN-synthesized or RTC-replaced is universally excluded as an independent detector measurement, even when finite. Synthesis or replacement elsewhere in its transitive support remains a mandatory cause-preserving influence input, not a universal RTC eligibility decision. Each downstream consumer owns the declared policy that maps that influence, response, and support to its own eligibility.

SCI-RTC-DEF-014 – Coordinate-dependent control.

A source mask or response operation that consumes an AST coordinate identity, frame, topology, detector binding, validity, uncertainty where required, and support. Invalid or unavailable coordinates are not outside-source values.

SCI-RTC-DEF-015 – Fixed sampling mode.

A request whose admitted factor, realizable prefilter, rate, phase-zero rule, and downstream compatibility are resolved without learning them from the observation.

SCI-RTC-DEF-016 – Learned sampling mode.

An optional request that evaluates an admitted finite candidate set and resolves one immutable observation-wide cadence/filter plan before apply. Its admission uses the smallest admitted beam, maximum valid in-scan speed, analytical beam-times-realized-filter response, exact phase-zero alias accounting, sampling sufficiency, filter realizability, and downstream compatibility. Noise-aware, per-array, per-scan, heterogeneous-cadence, and continuous adaptation are outside v0.1.

SCI-RTC-DEF-017 – Conditional statistical covariance.

The covariance obtained from an admitted input covariance through a fixed realized operator while holding selection and nuisance state fixed. It is not total uncertainty and does not imply a diagonal, white, stationary, or independent noise model.

SCI-RTC-DEF-018 – Atomic RTC bundle.

One role-specific completion unit containing conditioned raw x TOD; the immutable paired **rs** parent and pairing/mapping identity; every **rs**-derived diagnostic, selector, segmentation, and resolved-plan use; ordered detector/sample/scan/support and parent identity; complete response or typed unavailability; cause-preserving flags and influence; separated validity and eligibility inputs; conditional uncertainty/covariance status; one-way state and provenance; required diagnostics; downstream CAL handoff state; and completion/failure state. TOD alone is not this bundle.

SCI-RTC-DEF-019 – Scientifically named diagnostic.

A quantity with an explicit estimator, unit, support, selection/mask, validity, uncertainty/availability, and policy role. It is either inert, advisory, or an input to an explicitly selected policy; those roles are not interchangeable.

SCI-RTC-DEF-020 – Claim layer.

One of algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, or production readiness. Evidence at one layer does not establish another.

SCI-RTC-DEF-021 – Declared learning population.

The exact training interval, designated same-observation interval, scan collection, observation aggregate, or other named population used to infer an RTC quantity, including identity, support, selection, masks, exclusions, signal role, compatibility domain, and completeness. A population is not defined by whatever samples happen to be available during execution.

SCI-RTC-DEF-022 – Learned evidence.

The output of named diagnostics applied to a declared learning population, including candidate parameters, estimator, unit and convention, support, quality, uncertainty, validity, and failure state. Learned evidence proposes or constrains a plan; it does not authorize apply.

SCI-RTC-DEF-023 – Resolved RTC plan.

One immutable observation- and role-bound specification admitted only after all required scientific predicates pass. It contains operation sequence, family, exact coefficients or identity, rates and conventions, order/taps, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation and timing, expected response/support, uncertainty, validity, and downstream compatibility.

SCI-RTC-DEF-024 – Apply state.

Execution of exactly one resolved plan on compatible admitted input. Apply records actual intervals, exceptions, resets, outputs, and completion but does not relearn, retune, change policy, or rewrite the resolved plan.

SCI-RTC-DEF-025 – Online-adaptive estimator.

A separate time-varying, data-dependent estimator whose coefficients or other operator parameters evolve during apply under a declared update rule. Its learning support, stability, response, uncertainty, guard, reset, and restart semantics are not those of learned-then-fixed filtering. It is unavailable in v0.1 absent successor authority.

SCI-RTC-DEF-026 – Scientific filter design specification.

The physical purpose, source/contaminant model, admitted scan/beam/source domain, pass/transition/stop or notch constraints, phase and registration, alias bound, support/guard, numeric precision, realizability, validation, and failure criteria that justify a filter family and its numerical parameters. An operation label or coefficient array alone is not this specification.

SCI-RTC-DEF-027 – Bounded iterative notch-plan refinement.

A finite owner-bounded sequence of refinement attempts. Each attempt learns from a declared population and the current accepted-plan evaluation product, resolves a complete cumulative proposal and typed disposition, and advances the accepted-plan index only when that proposal is accepted. Apply of every accepted plan is immutable; rejected or stop attempts do not manufacture a new accepted plan or evaluation product. This is not the online-adaptive estimator of SCI-RTC-DEF-025.

SCI-RTC-DEF-028 – Complete cumulative successor proposal.

The full plan proposed by attempt a and accepted as Π_{k+1} only after its admission predicates pass, containing every retained, modified, removed, and rejected notch disposition; every broad-band stage; exact order, coefficients, precision, state, masks, edges, guards, intervals, decimation, timing, cumulative response/support, uncertainty, validity, and downstream compatibility. “Add this notch to the prior output” is not a complete successor plan.

SCI-RTC-DEF-029 – Final accepted plan and termination disposition.

The last complete plan whose admission predicates passed, together with one typed finite stop cause. No-successor, residual-satisfied, stable-plan, budget/stability/support/compatibility rejection, insufficient evidence, maximum-attempt, oscillating-plan, nonconvergent-parameter, and owner-cancelled states remain distinct. Maximum-attempt termination is not convergence.

SCI-RTC-DEF-030 – Paired raw x, r detector coordinates.

The two raw coordinates $\mathbf{u}_d = (x_d, r_d)^T$ produced for one exact detector occurrence by an admitted Tune-dependent readout mapping. The x coordinate carries the ordinary total-intensity analysis path; the nominally orthogonal r coordinate carries readout, leakage, artifact, and Tune-health information. Pairing does not assert equal scaling, response, validity, or uncertainty.

SCI-RTC-DEF-031 – Tune-dependent $I, Q \rightarrow x, r$ mapping authority.

The upstream authority identified by ζ that owns the microwave transmission coordinates, transform, reference point, normalization, sign, local Jacobian or nonlinear equivalent, validity/linearity domain, epoch, and uncertainty. RTC consumes its identity and admitted output but does not produce or repair the mapping.

SCI-RTC-DEF-032 – Optical leakage into the paired r coordinate.

A nonzero admitted optical-response projection in nominally orthogonal r . It is a readout-coordinate alignment or optical-leakage diagnostic, not automatically a literal Tune-angle error. A scalar ratio or angle tests only its declared optical projection and does not prove the complete transform.

SCI-RTC-DEF-033 – Atmospheric optical-leakage diagnostic.

A scientifically named estimate of paired x/r response to a declared network atmospheric optical-mode template, with exact membership, self-inclusion rule, time/frequency support, preprocessing, estimator, response, covariance, uncertainty, coherence, Tune/loading state, and scalar, frequency-dependent, or unavailable disposition.

SCI-RTC-DEF-034 – Bright-source optical-leakage diagnostic.

A separately parented estimate of paired x/r response to a declared localized source template and support, normally relevant to Beammap, Pointing, or OOF roles. It neither inherits nor silently replaces an atmosphere-derived diagnostic.

SCI-RTC-DEF-035 – Network-coherent detector-dependent level shift.

An unexpected persistent readout-baseline change having a common or near-common network change time and independently estimated detector/channel amplitudes $(\Delta x, \Delta r)$. Simultaneity, an r signature, or its absence does not by itself identify physical cause.

SCI-RTC-DEF-036 – Transition interval and guarded change-point support.

The accepted level-shift transition mask together with pre/post guards, timing uncertainty, affected detector population, and filter-state boundary. No FIR/IIR support may cross it absent separately authorized continuity.

SCI-RTC-DEF-037 – Stable plateau and role-specific admissibility.

A half-open interval between resolved transitions that is consistent with its declared signal-plus-baseline model, accounting for atmosphere, sources, noise, slow variation, Tune state, valid support, and absence of unresolved additional shifts. Suitability is role-specific and depends on the required estimator support, not sample count alone.

SCI-RTC-DEF-038 – Separately authorized two-channel correction or stitching estimator.

Any estimator that numerically mixes x and r , subtracts/rotates one into the other, or shifts plateau levels to manufacture continuity. It requires separate purpose, coupling and learning models, complete two-channel response and covariance, astronomical transfer, uncertainty, validation, restart, and provenance authority; none is a v0.1 default.

C Normative equations and identities

The equations in this section are the sole displayed mathematical authority of both views. They describe an implementation-independent, state-conditioned operator. They neither select numerical policy nor assert implementation conformity.

C.1 Raw RTC domain, donor transfer, and CAL handoff

For an exact detector occurrence i under the admitted multiplicative `flxscale` convention, define the detector-static comparison coordinate

$$\chi_i = \phi_i x_i, \quad \phi_i = \text{flxscale}_i. \quad (\text{SCI-RTC-EQ-001})$$

Equation (SCI-RTC-EQ-001) neither produces a calibrated RTC product nor assigns a role to legacy `responsivity`; target-atmosphere correction is absent. When q is a raw donor and d is a raw target, the unique multiplicative transfer that represents donor-calibrated amplitude in the target raw convention is

$$x_{d \leftarrow q}^R = a_{d \leftarrow q} x_q, \quad a_{d \leftarrow q} = \frac{\phi_q}{\phi_d}, \quad \chi_d^R = \phi_d x_{d \leftarrow q}^R = \phi_q x_q = \chi_q. \quad (\text{SCI-RTC-EQ-002})$$

This equation is defined only for a compatible `flxscale` pair as defined above. Otherwise this transfer is unavailable. An additive calibration convention, a different unit/reference domain, or an alternative donor authority requires its own complete affine equivalence and is not silently folded into equation (SCI-RTC-EQ-002).

Let $\Phi = \text{diag}(\phi_d)$ be the selected detector-static factor operator and $C_{\kappa, \mathcal{R}}^{\text{atm}}$ the target-atmosphere operator owned by SCI-CAL. The distinct downstream calibration operator is

$$C_{\kappa, \mathcal{R}}^{\text{CAL}} = \Phi C_{\kappa, \mathcal{R}}^{\text{atm}}, \quad \mathbf{c}_{\mathcal{R}}^{\text{CAL}} = C_{\kappa, \mathcal{R}}^{\text{CAL}} \mathbf{y}_{\mathcal{R}}^{\text{RTC}}. \quad (\text{SCI-RTC-EQ-003})$$

Equation (SCI-RTC-EQ-003) is an SCI-CAL handoff identity; its output is not `xs` and it is not an RTC stage. The selected cross-package order is

$$(\mathbf{x}^{\text{acq}}, \mathbf{r}^{\text{acq}}) \xrightarrow{A_\alpha, E_{\omega, \lambda}, F_1 \dots F_K, D_{M, 0}} \left(\mathbf{y}_{\mathcal{R}}^{x, \text{RTC}}, \mathbf{y}_{\mathcal{R}}^{r, \text{RTC}} \right) \xrightarrow{C_{\kappa, \mathcal{R}}^{\text{CAL}}} \mathbf{c}_{\mathcal{R}}^{\text{CAL}}. \quad (\text{SCI-RTC-EQ-004})$$

RTC shall not move the downstream CAL arrow inside replacement, temporal filtering, or phase-zero sampling. Only the x member is handed to SCI-CAL. The paired r member remains an RTC diagnostic coordinate and is not

calibrated, substituted for x , or interpreted as Stokes data. The raw donor ratio in equation (SCI-RTC-EQ-002) is convention transfer within $E_{\omega,\lambda}$, not absolute calibration. Any selected atmospheric template removal is part of $E_{\omega,\lambda}$ and its noncommutation with later filtering shall be represented by a response or declared bound.

C.2 Realized conditioning operator

Conditioned on complete realized state Ω , the ordered raw role-specific operator is

$$\mathbf{y}_{\mathcal{R}}^{xr, \text{RTC}} = D_{M,0} F_{K,\theta_K} \cdots F_{1,\theta_1} E_{\omega,\lambda,\mathcal{R}} (A_{\alpha} \mathbf{u}^{xr, \text{acq}} + \mathbf{a}_{\alpha}) + \mathbf{b}_{\Omega}. \quad (\text{SCI-RTC-EQ-005})$$

Chronological order is read from right to left. The affine terms include only declared replacement, filter-state, boundary, or other authorized contributions. Masks, coefficients, finite handling, context, exceptions, and state are components of Ω , not annotations outside the operator. Conditional on them,

$$\mathbf{y}_{\mathcal{R}}^{xr, \text{RTC}} = L_{\Omega} \mathbf{u}^{xr, \text{acq}} + \mathbf{b}'_{\Omega}, \quad L_{\Omega} = D_{M,0} F_K \cdots F_1 E_{\omega,\lambda,\mathcal{R}} A_{\alpha}. \quad (\text{SCI-RTC-EQ-006})$$

This is a conditional affine identity, not an unconditional linear, stationary, unbiased, or Gaussian claim.

A realized replacement at target cell (d, j) may be written

$$u_{dj}^R = (1 - p_{dj}) u_{dj} + p_{dj} \left[\sum_{(q,\ell) \in \mathcal{Q}_{dj}} w_{dj,q\ell} a_{d \leftarrow q,j\ell} u_{q\ell} + \beta_{dj} \right], \quad \sum_{q,\ell} w_{dj,q\ell} = 1 \text{ when normalization is claimed.} \quad (\text{SCI-RTC-EQ-007})$$

The scale a is domain-qualified: equation (SCI-RTC-EQ-002) is the authorized raw `flxscale` route. Any other transfer carries separate identity and validity and does not import downstream calibration into RTC.

For a time-invariant FIR stage,

$$v_{d,j} = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} u_{d,j-\ell}, \quad H_d^{(k)}(e^{i\omega}) = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} e^{-i\omega\ell}, \quad H_d^{(k)}(1) = \sum_{\ell} h_{d,\ell}^{(k)}. \quad (\text{SCI-RTC-EQ-008})$$

For a causal IIR or notch stage, a state-complete form is

$$\mathbf{s}_{j+1} = A_k \mathbf{s}_j + B_k u_j, \quad v_j = C_k \mathbf{s}_j + D_k u_j, \quad \mathbf{s}_{j_0} = \mathbf{s}_{0,k}. \quad (\text{SCI-RTC-EQ-009})$$

An exactly equivalent recurrence is admissible only with its coefficient convention, precision, state, and boundary identity. A causal-stability claim requires every realized pole strictly inside the unit circle.

One possible selected mask operator is normalized masked FIR convolution,

$$v_j = \frac{\sum_{\ell} h_{\ell} m_{j-\ell} u_{j-\ell}}{\sum_{\ell} h_{\ell} m_{j-\ell}}, \quad \left| \sum_{\ell} h_{\ell} m_{j-\ell} \right| > \epsilon_m, \quad (\text{SCI-RTC-EQ-010})$$

but equation (SCI-RTC-EQ-010) is not a selected default. Segmenting, zero filling, restoring a mask after filtering, holding state, and renormalizing are distinct operators.

C.3 Phase-zero sampling and response

For a coherent pre-sampling segment of length N , v0.1 selects only

$$y_{d,n}^{\text{RTC}} = v_{d,Mn}^{\text{preD}}, \quad 0 \leq Mn < N, \quad f_{\text{out}} = \frac{f_{\text{in}}}{M}, \quad N_y = \begin{cases} 0, & N = 0, \\ 1 + \left\lfloor \frac{N-1}{M} \right\rfloor, & N > 0. \end{cases} \quad (\text{SCI-RTC-EQ-011})$$

The representative assigned-grid time is the selected point time $t_n^{\text{out}} = t_{Mn}$; the scientific support is nevertheless the complete transitive support $\mathcal{S}_{dn}$ of all preceding stages. A selected edge/output policy may reject candidate points, but shall not relabel a different phase or silently change equation (SCI-RTC-EQ-011). The representative

source occurrence is exactly $\rho_{dn} = (d, Mn)$ in the pre-replacement acquired stream. It is not the largest response coefficient, a support centroid, a donor occurrence, or a synthetic sample.

For output (d, n) and acquired input (q, j) , the exact local response is

$$K_{dn,qj} = \frac{\partial y_{dn}}{\partial x_{qj}^{\text{acq}}} \quad \text{at fixed realized state } \Omega. \quad (\text{SCI-RTC-EQ-012})$$

At a selection boundary the derivative may be discontinuous or undefined; that state shall be represented rather than replaced by an LTI claim.

In the fixed, detector-separable, translation-invariant, single-rate interior case only,

$$H_d^{\text{comb}}(e^{i\omega}) = \prod_{k=1}^K H_d^{(k)}(e^{i\omega}), \quad \tau_g(\omega) = -\Delta t \frac{d}{d\omega} \text{unwrap}[\arg H_d^{\text{comb}}(e^{i\omega})]. \quad (\text{SCI-RTC-EQ-013})$$

Amplitude is $|H|$, power response is $|H|^2$, and phase is $\arg H$ under the recorded Fourier convention. Group delay is unavailable at response zeros unless a bounded convention is supplied.

For an LTI prefilter followed by phase-zero selection, the exact multirate identity is

$$Y(e^{i\Omega_o}) = \frac{1}{M} \sum_{p=0}^{M-1} H(e^{i(\Omega_o+2\pi p)/M}) X(e^{i(\Omega_o+2\pi p)/M}), \quad (\text{SCI-RTC-EQ-014})$$

under the declared Fourier/sign convention. Every folded image contributes; point selection itself is not anti-alias filtering.

For an FIR output, direct support starts with

$$\mathcal{S}_{dn}^{(1)} = \{Mn - \ell : \ell \in \mathcal{L}\}, \quad \mathcal{S}_{dn} = \text{closure}_{A,B,F,\text{state}}(\mathcal{S}_{dn}^{(1)}). \quad (\text{SCI-RTC-EQ-015})$$

IIR support extends to the declared state/reset or context boundary and may be mathematically infinite. Any finite guard is an approximation with a stated residual-tail criterion.

C.4 Moments, influence, and learned-plan selection

For admitted input mean $\boldsymbol{\mu}_x$ and full covariance Σ_x , conditional on fixed Ω independent of the modeled random noise,

$$\mathbb{E}[\mathbf{y} \mid \Omega] = L_\Omega \boldsymbol{\mu}_x + \mathbf{b}'_\Omega, \quad (\text{SCI-RTC-EQ-016a})$$

$$\Sigma_{y|\Omega}^{\text{stat}} = L_\Omega \Sigma_x L_\Omega^\top. \quad (\text{SCI-RTC-EQ-016b})$$

If selectors depend on the random data, unconditional covariance obeys

$$\text{Cov}(\mathbf{y}) = \mathbb{E}_\Omega[\text{Cov}(\mathbf{y} \mid \Omega)] + \text{Cov}_\Omega(\mathbb{E}[\mathbf{y} \mid \Omega]). \quad (\text{SCI-RTC-EQ-017})$$

The second term is not zero by default. With named nuisance parameters $\boldsymbol{\eta}$ and an adequate local linearization,

$$\Sigma_y^{\text{sys}} \approx J_\eta \Sigma_\eta J_\eta^\top, \quad J_\eta = \left. \frac{\partial \mathbb{E}[\mathbf{y} \mid \Omega, \boldsymbol{\eta}]}{\partial \boldsymbol{\eta}} \right|_{\boldsymbol{\eta}_0}, \quad (\text{SCI-RTC-EQ-018})$$

and a total-covariance claim requires every applicable component and cross term:

$$\Sigma_y^{\text{total}} = \Sigma_y^{\text{stat}} + \Sigma_y^{\text{sys}} + \Sigma_y^{\text{selection}} + \Sigma_y^{\text{model}} + \Sigma_y^{\text{cross}}. \quad (\text{SCI-RTC-EQ-019})$$

Unknown components are unavailable, not zero.

Define cause-preserving influence, direct exclusion, and consumer eligibility by

$$\mathcal{I}_{dn} = \text{closure}_{K, \text{generative links}}\{\text{typed source causes on } \mathcal{S}_{dn}\}, \quad (\text{SCI-RTC-EQ-020a})$$

$$E_{dn}^{\text{RTC,direct}} = V_d^{\text{det}} \wedge V_{dn}^{\text{num}} \wedge V_{dn}^{\text{support}} \wedge V_{dn}^{\text{response}} \wedge \neg D_{dn}^{\text{ALIGN_synth}} \wedge \neg D_{dn}^{\text{RTC_replace}},$$

$$E_{dn}^{(c)} = E_{dn}^{\text{RTC,direct}} \wedge P_c(\mathcal{I}_{dn}, \mathcal{S}_{dn}, K_{dn}, \dots) \quad (\text{SCI-RTC-EQ-020b})$$

D_{dn} identifies whether the exact representative occurrence ρ_{dn} was synthesized or replaced before subsequent filtering and selection. P_c is the declared policy of consumer c for noncenter transitive influence. RTC preserves every cause and numerical dependency but does not replace P_c with a universal transitive-ineligibility decision.

For learned mode, let $\mathcal{M}$ be the admitted finite integer candidate set and $\mathcal{F}(M)$ the conjunction of all owner-resolved astronomical transfer, exact phase-zero alias, sampling, realized-filter realizability, and downstream-compatibility constraints, evaluated with the smallest admitted beam and maximum valid in-scan speed. Then

$$\mathcal{M}_{\text{safe}} = \{M \in \mathcal{M} : \mathcal{F}(M) = \text{true}\}, \quad M_{\star} = \max \mathcal{M}_{\text{safe}}. \quad (\text{SCI-RTC-EQ-021})$$

Equation (SCI-RTC-EQ-021) is defined only after every numerical tolerance, support rule, fallback, and compatibility predicate named in the owner ledger is resolved. Percentile speed is diagnostic, not safety authority.

Finally, the atomic product is

$$\mathcal{O}_{\text{RTC}, \mathcal{R}} = (\mathbf{y}^x, \mathbf{y}^r, \mathcal{J}^{xr}, \mathcal{K}^{xr}, \mathcal{S}, \mathcal{I}, \mathcal{F}, \mathcal{V}, \mathcal{U}, \mathcal{P}, \mathcal{D}, \mathcal{C}), \quad (\text{SCI-RTC-EQ-022})$$

where $\mathcal{J}^{xr}$ is exact ordered pair identity, $\mathcal{K}^{xr}$ is complete paired response status, $\mathcal{S}$ support, $\mathcal{I}$ influence, $\mathcal{F}$ typed flags/causes, $\mathcal{V}$ separated validity and eligibility inputs, $\mathcal{U}$ uncertainty/covariance availability, $\mathcal{P}$ one-way state/provenance, $\mathcal{D}$ diagnostics, and $\mathcal{C}$ atomic completion or required-output failure state.

C.5 Scientific design and learn–resolve–apply identities

For a locally Gaussian effective beam with angular covariance Σ and angular scan-velocity vector $\mathbf{v}$, the projected source-crossing time scale is

$$\sigma_t = (\mathbf{v}^\top \Sigma^{-1} \mathbf{v})^{-1/2}. \quad (\text{SCI-RTC-EQ-023})$$

Equation (SCI-RTC-EQ-023) is a local Gaussian diagnostic. Scientific admission shall use the exact source-times-realized-response model over the declared beam, scan, source-class, and cadence domain.

A generic realized notch may be represented, where applicable, by

$$H_{\text{notch}}(z) = g \frac{\prod_{a=1}^{N_z} (1 - z_a z^{-1})}{\prod_{b=1}^{N_p} (1 - p_b z^{-1})}, \quad |p_b| < 1 \text{ for a causal-stability claim.} \quad (\text{SCI-RTC-EQ-024})$$

This form imposes no universal biquad. Exact realized coefficients, sample rate, frequency convention, phase/direction, state, edges, and precision are part of the response.

For an admitted FIR design family H_N , a selected order shall follow a declared constrained problem of the form

$$N_{\star} = \min_N \left\{ N : \begin{array}{l} \sup_{f \in \mathcal{P}} |H_N(f) - H_{\text{target}}(f)| \leq \epsilon_{\mathcal{P}}, \\ \sup_{f \in \mathcal{S}} |H_N(f)| \leq \epsilon_{\mathcal{S}}, \\ \text{phase/delay, alias, edge/guard, precision, and realizability predicates pass} \end{array} \right\}. \quad (\text{SCI-RTC-EQ-025})$$

The sets, metrics, tolerances, equality conventions, and family are owner-resolved; an approximate order formula is not universal authority.

For a causal linear-phase FIR with N taps at rate f_{samp} , its nominal interior group delay is

$$\tau_g = \frac{N - 1}{2f_{\text{samp}}}. \quad (\text{SCI-RTC-EQ-026})$$

Where a temporal response has effective delay τ during a locally uniform scan, the corresponding scan-direction angular displacement is

$$\Delta \boldsymbol{\theta} = \mathbf{v} \tau. \quad (\text{SCI-RTC-EQ-027})$$

For every learned RTC operation, the lifecycle is

$$\Lambda = \text{Learn}(\mathcal{L}; \Theta^{\text{eff}}), \quad \Pi = \text{Resolve}(\Lambda, \Theta^{\text{policy}}), \quad \mathbf{y} = \mathcal{A}_{\Pi}(\mathbf{x}), \quad \frac{\partial \Pi}{\partial j} = 0 \text{ during apply.} \quad (\text{SCI-RTC-EQ-028})$$

The final equality denotes plan immutability over apply slots; it is not an assumption that the realized operator is stationary at masks, edges, state boundaries, or other declared exceptions.

For bounded iterative notch-plan refinement with original admitted input $x^{(0)}$, initial fixed or identity plan Π_0 , actual attempt count A , configured maximum $A_{\max}$, attempt index $a \in \{1, \dots, A\}$, and accepted-plan index k_a available at the start of attempt a ,

$$\begin{aligned}
x^{\text{eval},(0)} &= \mathcal{A}_{\Pi_0}(x^{(0)}), & k_1 &= 0, \\
\Lambda_a &= \text{Learn}(\mathcal{L}_a, x^{\text{eval},(k_a)}, \Pi_{k_a}), \\
(\tilde{\Pi}_a, \delta_a) &= \text{Resolve}(\Lambda_a, \Pi_{k_a}, \Theta^{\text{policy}}), \\
k_{a+1} &= k_a + \mathbb{I}[\delta_a = \text{accepted}], \\
\delta_a = \text{accepted} &\implies \begin{cases} \Pi_{k_{a+1}} = \tilde{\Pi}_a, \\ x^{\text{eval},(k_{a+1})} = \mathcal{A}_{\Pi_{k_{a+1}}}(x^{(0)}) \end{cases}, & (\text{SCI-RTC-EQ-029}) \\
\delta_a \neq \text{accepted} &\implies \Pi_{k_{a+1}} = \Pi_{k_a} \quad \text{and no new evaluation is required,} \\
x^{\text{final}} &= \mathcal{A}_{\Pi_K}(x^{(0)}), & K = k_{A+1} \leq A \leq A_{\max} < \infty, \\
\frac{\partial \Pi_k}{\partial j} &= 0 \quad \text{during every apply of accepted plan } k.
\end{aligned}$$

Equation (SCI-RTC-EQ-029) separates the actual finite attempt count from its configured maximum and from the accepted-plan sequence. It does not create artificial no-op attempts after an earlier stop. It defines the initial evaluation for nonidentity Π_0 , and states the default original-input replay rule. A sequential cascade from a preceding evaluation product requires separate explicit operator authority and complete propagation of every intermediate parent, response, state, phase, edge, support, covariance, and uncertainty.

C.6 Paired coordinates, leakage, and level shifts

For every admitted detector occurrence, the upstream mapping authority shall produce one ordered pair from measured in-phase and quadrature coordinates,

$$\begin{bmatrix} x_{dj}^A \\ r_{dj}^A \end{bmatrix} = T_{dj}^{IQ \rightarrow xr} \begin{bmatrix} I_{dj}^A \\ Q_{dj}^A \end{bmatrix} + \mathbf{c}_{dj}^{xr}, \quad J_{dj}^{xr} = (d, j, \text{stream}, \text{mapping revision}). \quad (\text{SCI-RTC-EQ-030})$$

The pair identity is exact. Missing either member is malformed required input; validity may nevertheless differ between the two members.

An admitted local optical response model may be written

$$\begin{bmatrix} \Delta x_d \\ \Delta r_d \end{bmatrix} = \begin{bmatrix} \alpha_d^x \\ \alpha_d^r \end{bmatrix} s_d + \begin{bmatrix} \epsilon_d^x \\ \epsilon_d^r \end{bmatrix}, \quad \alpha_d^x \neq 0. \quad (\text{SCI-RTC-EQ-031})$$

This is diagnostic authority, not permission to correct one member with the other or to assume $\alpha_d^r = 0$.

For a declared source class c and support-qualified estimator,

$$\zeta_d^{(c)} = \frac{\alpha_d^{r,(c)}}{\alpha_d^{x,(c)}}, \quad \Delta r_d = \zeta_d^{(c)} \Delta x_d + \varepsilon_d^{(c)}. \quad (\text{SCI-RTC-EQ-032})$$

Atmosphere-derived and bright-source-derived estimates are distinct products with distinct parentage, support, validity, uncertainty, and scalar-versus- frequency-dependent status.

For a declared level-shift event e at transition τ_e , the paired piecewise-additive model is

$$u_{dj}^c = u_{dj}^{c,\text{base}} + \sum_{e \in \mathcal{E}_d} \Delta_{de}^c \mathbb{I}[j \geq \tau_{de}], \quad c \in \{x, r\}. \quad (\text{SCI-RTC-EQ-033})$$

The event time, both amplitudes or their unavailability, detection state, transition mask, guards, and resulting segment boundaries belong to the realized plan. Detection is distinct from correction.

Within a segment g delimited by admitted resets and level shifts, a plateau diagnostic and reset condition may be represented by

$$P_{dg}^c = \text{Plateau}(u_{d,g}^c; \Theta_P), \quad \mathbf{s}_{j_g^+} = R_g(\mathbf{s}_{j_g^-}; \Theta_R), \quad c \in \{x, r\}. \quad (\text{SCI-RTC-EQ-034})$$

No plateau value authorizes cross-segment stitching unless separately selected, propagated, and validated.

The actual bounded-refinement counts satisfy

$$0 \leq K \leq A \leq A_{\max} < \infty, \quad A = \#\{a : \text{Learn}_a \text{ was actually attempted}\}. \quad (\text{SCI-RTC-EQ-035})$$

D Bounded assumptions and non-goals

SCI-RTC-ASM-001 – Capability.

V0.1 admits the exact paired raw readout coordinates **xs** and **rs**. The x member is the ordinary science-conditioning path; the r member is a retained diagnostic coordinate. Raw x is not itself a Stokes observable, and enabled polarimetry remains excluded until separately approved.

SCI-RTC-ASM-002 – Aligned-grid boundary.

The input grid, cadence, phase, mapping, gaps, synthesis/origin state, support, and scan partition are admitted ALIGN facts. Physical event timing, absolute timing accuracy, and a producer epoch may remain unavailable.

SCI-RTC-ASM-003 – Coordinate boundary.

Coordinate-dependent operations assume an admitted AST identity, frame, topology, detector binding, support, validity, and uncertainty where required. The absence of such authority does not license inference or clamping.

SCI-RTC-ASM-004 – Calibration boundary.

Calibration-factor production, target-atmosphere correction, beam meaning, uncertainty, and promotion belong to SCI-BEAM/SCI-CAL. RTC remains in raw $\Delta f/f$ through replacement, filtering, and sampling. RTC may learn and apply a separately authorized raw atmospheric template and shall account for its noncommutation with later filters. SCI-CAL consumes only the exact RTC x product and applies absolute **flxscale** plus target atmosphere only after RTC.

SCI-RTC-ASM-005 – Selected-policy boundary.

Despite, level-shift detection, transition masks, segmentation, reset, plateau use, donor, mask, FIR, IIR, notch, atmospheric-template, leakage, state, edge, non-finite, recovery, sampling, diagnostic, and failure choices are selected versioned policies. No current default or numerical threshold is promoted to a universal scientific constant by this contract.

SCI-RTC-ASM-006 – Conditional linearity.

Linear response and covariance identities condition on fixed paired-stream selection and realized state. Cross-coordinate covariance and shared selectors are retained. Data-derived selectors make the unconditional operation generally nonlinear and nonstationary.

SCI-RTC-ASM-007 – Statistical input.

Moments or covariance are claimed only from a supplied statistical model with declared correlation and limitations. Absence does not imply diagonal, white, stationary, independent, or zero covariance.

SCI-RTC-ASM-008 – Learned-mode scope.

Every learned-then-fixed RTC operation uses one declared learning population, one resolved immutable plan, and one explicit compatibility domain. The first learned sampling scope produces one common observation plan. Per-array, per-scan, heterogeneous-cadence, noise-aware, and continuously adaptive sampling execution are successor scopes. Bounded iterative notch-plan refinement is admitted only as a finite outer sequence of actual attempts, bounded by a separately configured maximum, over complete immutable accepted plans; rejected attempts do not advance the accepted-plan sequence, unused maximum attempts are not artificial no-ops, and no parameter evolves within an apply pass.

SCI-RTC-ASM-009 – Analytical authority.

Exact analytical source-times-realized-filter transfer and phase-zero alias accounting define learned-plan admission. Deterministic vectors may falsify a realization, while source injections and observational studies are required for separately claimed performance or science-impact evidence.

SCI-RTC-ASM-010 – Compact representation.

Dense observation-sized response, covariance, or per-sample provenance is not required when a factorized, sparse, interval-based, or generative representation exactly reconstructs the operator, support, causes, and exceptions.

SCI-RTC-ASM-011 – Downstream authority.

RTC supplies causal inputs but does not own PTC cleaning/weights, VAL eligibility policy, MAP estimation, BEAM inference, empirical NOI products, MAP-003 response tracing, map filtering, source fitting, or fruit-loop recurrence.

SCI-RTC-ASM-012 – Claim separation.

The contract is an algebraic and semantic authority only. It contains no claim that a present implementation conforms, that a representation is faithful, that observational performance is adequate, or that a mode is ready for production.

D.1 Explicit non-goals

This revision does not derive ALIGN timing, AST coordinates, SCI-BEAM or SCI-CAL calibration, PTC, VAL, MAP, or FLT estimators, source fitting, or recurrence; authorize absolute-timing or astrometric corrections; redesign mature despiking/FIR/IIR/notch numerics; choose universal numerical thresholds; authorize online adaptation; activate polarimetry; authorize automatic r -to- x correction, r calibration, use of r as an x donor, or plateau-based cross-segment stitching; prescribe source classes, files, storage schemas, or product names; authorize per-array/per-scan cadence; inspect implementation; perform validation or reductions; or assert production status.

E Sequential normative requirements

The following identifiers are stable within v0.1/r0.5. Conformance is requirement-by-requirement; satisfying a neighboring requirement does not waive another one.

Area	ID	Normative statement
Boundary	SCI-RTC-REQ-001	RTC shall admit only exact aligned xs/rs pairs whose signals are raw, uncalibrated detector coordinates and whose ordered pair identity, assigned grid, support, origin/synthesis state, and independently declared validity are present; a missing partner is malformed required input.
Boundary	SCI-RTC-REQ-002	Every requested RTC product role shall declare the paired x/r sign, reference/baseline, valid domain, plan, and required consumers while retaining raw coordinate identity; no role shall give either member a calibrated or Stokes meaning.
Boundary	SCI-RTC-REQ-003	Every RTC conditioned detector output, including Beammap, Pointing, OOF, Science, and diagnostic-only roles, shall preserve the paired raw x/r coordinates and shall not be silently replaced by a calibrated timestream or map; the r member is diagnostic rather than an alternative science signal.
Boundary	SCI-RTC-REQ-004	A calibrated detector signal shall be a distinct downstream SCI-CAL product in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention, with admitted factor, atmosphere, detector, validity, uncertainty, and provenance state.
Boundary	SCI-RTC-REQ-005	RTC shall preserve the exact paired raw parent, plan, response, support, factor authority, RTC atmospheric-template lineage, and SCI-CAL atmosphere lineage required by downstream composition; it shall represent or bound noncommutation between RTC template removal and filtering but shall not apply, derive, repair, promote, or reinterpret SCI-CAL calibration.
Identity	SCI-RTC-REQ-006	Observation, Tune, array, network/interface, detector occurrence, APT binding, native row, aligned slot, scan, science interval, context interval, output slot, stage, and product identity shall remain distinct and immutable.
Identity	SCI-RTC-REQ-007	The ordered detector and sample shapes, index bases, native-to-aligned mapping, aligned-to-output map, scan membership, and memory order shall be explicit; indices shall not serve as cross-artifact detector identity.
Identity	SCI-RTC-REQ-008	RTC shall preserve the admitted ALIGN cadence, phase, rate, gaps, support, and synthesis/origin state without claiming physical integration-event timing or absolute epoch authority.
Identity	SCI-RTC-REQ-009	A coordinate-dependent operation shall consume the exact AST identity, frame, topology, detector binding, support, and validity required by its product role, with no implicit frame conversion.
State	SCI-RTC-REQ-010	Requested, effective, observation-resolved, learned/resolved when applicable, and realized state shall be recorded separately in one-way order; later state shall not overwrite the accepted request.
State	SCI-RTC-REQ-011	Outer, inner, full, mini, diagnostic, simulated, processed, and any other produced stage roles shall have distinct immutable identities and explicit parent/processing links without requiring duplicate computation.
Policy	SCI-RTC-REQ-012	Every despike, donor, replacement, mask, filter, state, edge, non-finite, recovery, sampling, flag, diagnostic, and fallback action shall be governed by an admitted versioned selected policy rather than an undocumented default.
Order	SCI-RTC-REQ-013	RTC shall complete paired raw-domain alignment, selected early despike/level-shift composition, compatible x donor transfer/replacement, selected atmospheric-template removal, temporal filtering, and phase-zero sampling before SCI-CAL consumes only x and applies absolute flxscale and target-atmosphere correction as in equation (SCI-RTC-EQ-004); no calibrated RTC branch or alternative interleaving is authorized in v0.1.

Area	ID	Normative statement
Donor	SCI-RTC-REQ-014	A raw donor q to raw target d using the detector-static comparison coordinate $\chi_i = \phi_i x_i$ shall use ϕ_q/ϕ_d only when both flxscale factors are valid for the exact occurrences under the same compatible convention/domain and $\phi_d \neq 0$.
Donor	SCI-RTC-REQ-015	The raw donor route in REQ-014 shall use flxscale and shall neither require nor restore a scientific role for legacy responsivity .
Donor	SCI-RTC-REQ-016	When REQ-014 is not available, replacement shall fail under that route unless a separate donor-to-target authority declares direction, quantity/unit domain, support, uncertainty, validity, and provenance.
Donor	SCI-RTC-REQ-017	The despite detector, target indicator, candidate eligibility, topology, score, tie rule, donor weights, time relation, source-mask/flag use, fallback, and every realized donor link shall be reconstructible.
Donor	SCI-RTC-REQ-018	A replaced target shall retain target identity, link every donor and factor, preserve the replacement cause, and expose correlations from reused or shared donors; finiteness shall not create independent exposure.
Influence	SCI-RTC-REQ-019	An output whose exact representative source occurrence $\rho_{dn} = (d, Mn)$ is ALIGN-synthesized shall be universally excluded as an independent detector measurement; nonrepresentative synthesized influence shall preserve its numerical effect, typed cause, and support for the downstream consumer's declared eligibility policy.
Influence	SCI-RTC-REQ-020	An output whose exact representative source occurrence $\rho_{dn} = (d, Mn)$ is RTC-replaced shall be universally excluded as an independent detector measurement; nonrepresentative replacement influence shall preserve its numerical effect, typed cause, donor linkage, and support for the downstream consumer's declared eligibility policy.
Filter	SCI-RTC-REQ-021	Every enabled response-changing stage shall record chronological order, exact full-precision coefficients or exact digest, coefficient convention, normalization, numeric precision, rate, direction, state, context, mask, edge, non-finite policy, exceptions, and the coherent segment bounded by gaps, resets, or level shifts.
Filter	SCI-RTC-REQ-022	An FIR claim shall expose exact impulse support, DC gain, phase convention, and normalization; a label such as "normalized" shall not substitute for the coefficient sum or response.
Filter	SCI-RTC-REQ-023	An IIR or notch claim shall expose an exact recurrence or state-space equivalent, realized poles/zeros where applicable, initial/final state, reset/carry boundary including admitted level shifts, transient/guard rule, precision, and stability status; state shall not silently cross a declared segment boundary.
Mask	SCI-RTC-REQ-024	A processing mask shall be represented as an operator control, including geometry or selection, boundary convention, dilation, denominator/segment behavior, and state interaction; it shall not be treated as acquisition validity or confidence.
Mask	SCI-RTC-REQ-025	Invalid, unavailable, wrong-frame, out-of-domain, or ambiguously bound coordinates shall not become outside-source samples and shall fail the affected required coordinate operation.
Finite	SCI-RTC-REQ-026	A non-finite required sample, factor, coefficient, coordinate, or filter state shall be rejected, marked unavailable over its exact transitive influence, or handled by an explicitly authorized prior replacement/recovery rule; it shall not be silently coerced to zero.
Edge	SCI-RTC-REQ-027	Boundary extension, incomplete context, filter guards, IIR recovery, short/empty scans, and output selection shall be explicit; an edge or incomplete-state cell shall not be labeled an ordinary interior sample.

Area	ID	Normative statement
Sampling	SCI-RTC-REQ-028	V0.1 downsampling shall use only phase-zero point selection $y^{\text{RTC}}[n] = v^{\text{preD}}[Mn]$ from the final pre-decimation stream; the representative source occurrence is exactly (d, Mn) before replacement/filtering, and arithmetic-mean or other block-aggregate downsampling is excluded.
Sampling	SCI-RTC-REQ-029	Each coherent segment shall record M , zero phase, input/output rates and grids, the cardinality of equation (SCI-RTC-EQ-011), selected-point representative time, full transitive support, edge disposition, and flag/influence aggregation.
Sampling	SCI-RTC-REQ-030	Any alias-free or bounded-alias claim shall include the exact realized prefilter and all folded bands of equation (SCI-RTC-EQ-014), with a declared input band, metric, bound, and unavailable state; point selection alone shall not be called anti-alias filtering.
Sampling	SCI-RTC-REQ-031	Fixed mode shall resolve an admitted factor/filter/rate/compatibility plan before application and record it independently of realized output.
Learned	SCI-RTC-REQ-032	Every learned RTC operation shall preserve distinct requested, effective, observation-resolved, learned-evidence, resolved-plan, and applied states and shall not advertise a plan before admitted evidence and numeric policy are complete.
Learned	SCI-RTC-REQ-033	Learned resolution shall choose the largest admitted integer factor only after astronomical-transfer, exact phase-zero alias, sampling, filter-realizability, and downstream-compatibility constraints all pass using the smallest admitted beam and maximum valid in-scan speed.
Learned	SCI-RTC-REQ-034	The first learned applied scope shall use one common immutable observation cadence and filter across arrays and scans; percentile speed shall remain diagnostic rather than safety authority.
Learned	SCI-RTC-REQ-035	Apply shall consume exactly one immutable resolved plan and shall not retune, adapt, or silently fall back; an unavailable or rejected plan shall follow the selected failure/fallback policy before apply begins.
Learned	SCI-RTC-REQ-036	Restart shall restore a state-complete resolved plan, verify exact plan/input compatibility, and reject stale or incompatible beam, speed, scan, coefficient, rate, phase, role, or parent identity.
Response	SCI-RTC-REQ-037	The atomic bundle shall contain the complete realized response of SCI-RTC-DEF-010 or a typed unavailable state; a partial filter kernel shall not be labeled the conditioned response.
Response	SCI-RTC-REQ-038	Selection, detector mixing, masking, finite boundaries, state, or multirate behavior shall use a local/factorized detector-time response or truthful unavailability rather than an unjustified scalar transfer; downstream SCI-CAL response remains a separate composition.
Response	SCI-RTC-REQ-039	A scalar LTI response shall be claimed only on a stated uniform, fixed-state, detector-separable, translation-invariant interior domain with no data-derived selector or undeclared rate change.
Response	SCI-RTC-REQ-040	Where defined, response shall distinguish amplitude, power, phase, group delay, sampling rate, Fourier convention, alias images, support, and response uncertainty; undefined phase/delay at a response zero shall remain unavailable.
Response	SCI-RTC-REQ-041	Output support shall be expanded transitively through alignment, replacement, every filter, state boundary, edge rule, and phase-zero selection; a selected center index shall not stand in for full support.

Area	ID	Normative statement
Statistics	SCI-RTC-REQ-042	Conditional mean and covariance shall follow equations (SCI-RTC-EQ-016a)–(SCI-RTC-EQ-016b) only for an admitted joint x/r input model and fixed realized state, with cross-coordinate covariance, reused donors, shared selectors, and overlapping support represented as correlations.
Statistics	SCI-RTC-REQ-043	When selection depends on modeled random data, selection bias and selection covariance shall be modeled or explicitly unavailable; they shall not be set to zero merely by conditioning on the realized selector.
Statistics	SCI-RTC-REQ-044	Calibration, atmosphere, beam, donor-scale, coefficient, cadence/phase, response, and model nuisance terms shall retain value, validity, uncertainty, provenance, and correlation scope, including required cross terms.
Statistics	SCI-RTC-REQ-045	Unknown covariance shall remain unavailable and shall not be replaced by diagonal, white, stationary, independent, zero, or scalar-weight assumptions; conditional inverse variance shall not be called full precision or total uncertainty.
Validity	SCI-RTC-REQ-046	Acquisition support, independent x validity, independent r validity, pair identity, ALIGN synthesis, numerical computability, coordinate validity, flags/causes, operator masks, replacement, transitive influence, complete support, response status, consumer eligibility, uncertainty availability, and provenance shall remain separate states.
Validity	SCI-RTC-REQ-047	A flag shall describe a cause and shall not automatically become a mask, invalidity, weight, or universal eligibility decision; aggregation shall preserve every contributing cause required by the selected policy.
Output	SCI-RTC-REQ-048	RTC shall produce the role-specific paired atomic bundle of equation (SCI-RTC-EQ-022); conditioned x TOD alone or conditioned r diagnostics alone shall not be reported as a complete required RTC product.
Output	SCI-RTC-REQ-049	Required malformed identity/state or missing authority shall propagate to atomic required-output failure; optional unavailable provenance detail shall be typed and numerically inert.
Output	SCI-RTC-REQ-050	Compact provenance shall reconstruct the exact IQ-to- x/r mapping and pair identity, input/parent identity, detector binding, grid and intervals, downstream CAL handoff authority, donor links, level shifts/segments/resets, ordered filters, masks, coefficients/state/edges, sampling, response/uncertainty status, exceptions, counts, and all one-way state records.
Output	SCI-RTC-REQ-051	Each diagnostic, including leakage and level-shift products, shall name estimator, unit, support, mask/selection, validity, availability, scalar-versus-frequency status, and whether it is inert, advisory, or enters the selected policy; optional diagnostic detail shall not change values, flags, eligibility, cardinality, or failure.
Consumer	SCI-RTC-REQ-052	Beyond the direct exclusions in REQ-019–020, RTC shall supply cause-preserving transitive influence rather than a universal downstream eligibility bit; each consumer shall bind the exact RTC parent, declare its influence-to-eligibility policy, and shall not erase causes or strengthen unavailable timing, coordinate, calibration, response, covariance, or eligibility claims.
Consumer	SCI-RTC-REQ-053	PTC-disabled terminal RTC output shall be a valid requested state when selected; no PTC or map product shall be implied, while any requested downstream product shall require the exact admitted RTC bundle.
Claims	SCI-RTC-REQ-054	Algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, and production readiness shall be stated and assessed as separate claims.

Area	ID	Normative statement
Lifecycle	SCI-RTC-REQ-055	Every learned operation shall name its learning population, estimator, support, selection, exclusions, quality, uncertainty, validity, and compatibility domain; same-observation learning shall be identified as data-dependent.
Lifecycle	SCI-RTC-REQ-056	Resolve shall convert evidence and fixed policy into one immutable plan containing every applicable sequence, family, coefficient identity, rate/convention, order/taps, band/notch parameter, applicability, mask, donor, edge/guard/reset, decimation/timing, response/support, uncertainty, validity, and downstream-compatibility field.
Lifecycle	SCI-RTC-REQ-057	A plan shall be resolved only when every required predicate and tolerance passes; missing evidence, missing tolerance, rejected evidence, or unavailable compatibility shall not be treated as pass.
Lifecycle	SCI-RTC-REQ-058	Any coefficient or operator parameter that evolves during apply shall be governed by a separately authorized online-adaptive estimator contract; it shall not be represented as learned-then-fixed filtering.
Design	SCI-RTC-REQ-059	Every despoke, donor, mask, notch, low/high/band-pass, FIR, IIR, decimation, edge/state, or non-finite operation shall declare scientific purpose, signal/contaminant model, mathematical operator, design parameters, learn/resolve/apply behavior, astronomical and calibration consequences, validation/falsification, and unavailable states.
Design	SCI-RTC-REQ-060	Filter admission shall use the exact source-times-realized-response over the declared scan-speed/direction, projected-beam, source-class, prior-response, and cadence domain; $f \sim v/\theta$, a scalar FWHM, or a nominal cutoff shall remain pedagogical only.
Notch	SCI-RTC-REQ-061	Every notch shall resolve center, rate/frequency convention, width or Q, depth, family/order, poles/zeros where applicable, phase/group delay, state, transient/guard, applicability, learned uncertainty, drift tolerance, and combined response for overlaps.
Bands	SCI-RTC-REQ-062	Every admitted low-, high-, or band-pass shall resolve pass/transition/stop bands, ripple, attenuation, phase/delay, family, order/taps, precision, edges/guards, retained astronomical modes, and unavailable modes; no class label shall substitute for exact response.

Area	ID	Normative statement
FIR	SCI-RTC-REQ-063	FIR order or tap count shall follow owner-resolved transfer, alias, phase, support, guard, precision, and realizability constraints such as equation (SCI-RTC-EQ-025); an arbitrary integer or approximate family formula shall not be scientific authority.
Donor	SCI-RTC-REQ-064	Donor replacement shall be identified as a continuity surrogate at the target identity and shall separately establish numerical convention transfer, donor admissibility, and scientific eligibility; compatible <code>flxscale</code> shall not be evidence of common sky signal.
Donor	SCI-RTC-REQ-065	Beammap-role donor scaling shall use an explicitly identified prior APT or other pre-existing calibration authority and shall not use a new <code>flxscale</code> derived from the same Beammap conditioning lineage.
Sampling	SCI-RTC-REQ-066	Decimation factor and prefilter shall be justified from cadence, exact response, folded-band alias bounds, smallest projected crossing time, maximum valid scan speed, source class, timing/coordinate accuracy, and downstream compatibility.
Timing	SCI-RTC-REQ-067	Each plan shall resolve temporal registration through zero phase, shifted output time, coordinate correction, response-aware downstream use, or an approved bound on residual centroid/PSF effect; selected grid time alone shall not imply response centering.
CAL	SCI-RTC-REQ-068	Calibration transfer between Beammap and target processing shall bind the complete RTC plans and require matching response or explicit calculation, renormalization, or observational evidence; identical operation labels or preserved units are insufficient.
Validation	SCI-RTC-REQ-069	Validation of a notch, broad-band/FIR, donor, or decimation plan shall vary its scientific design parameters over the admitted scan/beam/source domain and measure astronomical transfer, phase/centroid, PSF/morphology, ringing, support/edge loss, covariance, and alias or interference residual as applicable.
Claims	SCI-RTC-REQ-070	No plan shall be called science-qualified solely because coefficient algebra, optimization, serialization, or deterministic operator fixtures pass.

Area	ID	Normative statement
Iteration	SCI-RTC-REQ-071	Iterative notch-plan refinement shall use a finite owner-approved maximum attempt bound and stopping policy; actual attempt count, configured maximum, and accepted-plan count shall remain distinct, unused maximum attempts shall not become artificial no-ops, and an unbounded repeat-until-satisfactory process is unavailable.
Iteration	SCI-RTC-REQ-072	Every apply of accepted plan Π_k shall consume exactly that immutable plan and shall not change coefficients, notch parameters, policy, masks, donor rules, decimation, or state conventions.
Plan	SCI-RTC-REQ-073	Every proposed successor $\tilde{\Pi}_a$ shall be a complete cumulative plan that dispositions all prior and proposed notches and records all stages, order, coefficients/precision, state, masks, edges/guards, intervals, decimation/timing, response/support, uncertainty, validity, and compatibility; rejection shall not assign it an accepted-plan index.
Replay	SCI-RTC-REQ-074	The default evaluation and final production rule shall apply each complete accepted plan to original admitted $x^{(0)}$ as in equation (SCI-RTC-EQ-029); exploratory attempts shall not accumulate hidden rounding, padding, state, duplicate-filter, mask, flag, or parent effects.
Cascade	SCI-RTC-REQ-075	A cascade from a preceding filtered product shall require explicit scientific-operator authority and immutable intermediate parents, with total response, state, phase, edges, support, covariance, uncertainty, and sequential parentage retained.
Evaluate	SCI-RTC-REQ-076	Each successor attempt shall learn from the current accepted-plan evaluation and assess intended line attenuation and residual, parameter error/drift, predicted-versus-measured response, stability, ringing/transient, edge/guard loss, phase/delay, source amplitude/centroid/PSF, extended transfer, covariance, and support where applicable.

Area	ID	Normative statement
Candidate	SCI-RTC-REQ-077	A successor line shall be admitted only after comparison of original, predicted, and conditioned spectra, cumulative response, line uncertainty, astronomical spectrum, transfer/alias budgets, and detector/network/array/scan/interval consistency excludes notch-edge, ringing, transient, leakage, normalization, numerical, and response-floor artifacts.
Budget	SCI-RTC-REQ-078	Successor admission shall preserve owner-approved cumulative interference improvement, removed bandwidth, astronomical transfer, source amplitude/centroid/PSF, phase/registration, ringing/transient, guard/support, stability, cadence/response compatibility, uncertainty, and provenance budgets.
Stability	SCI-RTC-REQ-079	Plan stability shall compare the complete line set and matching identity, centers, widths/Q, depths, family/order, precision, applicability, state/reset, masks/guards, broad-band stages, decimation/timing, response/support, and compatibility over the required consecutive accepted-plan comparisons.
Stop	SCI-RTC-REQ-080	Every attempt shall record one typed accepted, rejected, or stop disposition plus its cause; maximum-attempt, oscillating-plan, and nonconvergent-parameter terminations shall not be labeled convergence or plan stability, and rejection shall leave the last accepted plan available unless policy says otherwise.
Lineage	SCI-RTC-REQ-081	Every attempt shall preserve attempt index, current accepted-plan index, original input, previous accepted plan, learning parent/population and PSD definition, original/predicted/measured spectra, proposal and disposition, predicates/tolerances, any newly accepted complete plan/evaluation product, cumulative response/astronomical diagnostics, stop cause, and final plan identity in one-way state.
Restart	SCI-RTC-REQ-082	Serialization and restart shall reproduce the final selected complete plan and declared parent rule exactly and reject incompatible original input, rate, learning population, parent, coefficient, interval, state, or policy identity.

Area	ID	Normative statement
Pair	SCI-RTC-REQ-083	Every required acquired occurrence shall contain both x_{dj}^A and r_{dj}^A with exact common detector, sample, stream, and mapping-revision identity; a missing member shall fail required-input admission rather than be synthesized, copied, or zero-filled.
Pair	SCI-RTC-REQ-084	Paired identity shall remain exact through selection, segmentation, filtering, sampling, serialization, and restart; reordering or independently joining the two members is not an equivalent representation.
Mapping	SCI-RTC-REQ-085	The upstream $IQ \rightarrow xr$ mapping shall expose authority, revision, coefficient/offset convention, applicability, validity, and provenance for each admitted occurrence; RTC shall consume rather than infer or repair that mapping.
Validity	SCI-RTC-REQ-086	Numerical and scientific validity of x and r shall be represented independently while retaining the pair; invalidity of one member shall not silently invalidate, replace, or validate the other.
Leakage	SCI-RTC-REQ-087	Optical response in r shall not be assumed zero. Any leakage claim shall identify the source class, estimator, support, selection, validity, uncertainty, and response convention.
Leakage	SCI-RTC-REQ-088	An atmosphere-derived x/r leakage diagnostic shall identify the atmospheric template or population, temporal/frequency support, masks, estimator, quality, uncertainty, and applicability independently of any bright-source result.
Leakage	SCI-RTC-REQ-089	A bright-source-derived x/r leakage diagnostic shall identify the source, model, spatial/temporal support, masks, estimator, quality, uncertainty, and applicability independently of any atmospheric result.
Leakage	SCI-RTC-REQ-090	Atmosphere-derived and bright-source-derived leakage products shall retain separate parentage and shall not be averaged, substituted, or promoted as mutually validating without an explicitly authorized comparison model.

Area	ID	Normative statement
Leakage	SCI-RTC-REQ-091	Each leakage product shall state whether it is scalar, band-limited, or frequency dependent and shall provide the corresponding response/support or a typed unavailable state.
Boundary	SCI-RTC-REQ-092	V0.1 shall not automatically correct x using r , calibrate r , use r as an x donor, or treat a valid r coordinate as a substitute for an invalid x coordinate.
Statistics	SCI-RTC-REQ-093	Any joint x/r selector, estimator, covariance, or diagnostic shall preserve cross-coordinate covariance, common support, shared selection, and unavailable components; independence shall require evidence.
Shift	SCI-RTC-REQ-094	Every level-shift candidate and disposition shall record detector, event/transition time or interval, x and r amplitudes or unavailability, estimator, support, quality, uncertainty, detection state, and cause.
Shift	SCI-RTC-REQ-095	The owner-selected early composition is despite detection and admitted sample replacement before level-shift estimation on the repaired paired stream; the exact policy revision and resulting masks shall be recorded, and no alternative suborder is implied.
Shift	SCI-RTC-REQ-096	A selected level shift shall create an explicit transition mask and owner-resolved pre/post guard whose samples remain distinguishable from ordinary segment interior.
Shift	SCI-RTC-REQ-097	A selected level shift shall delimit coherent processing segments and apply the selected reset/carry rule; filter, estimator, or donor state shall not silently cross that boundary.
Plateau	SCI-RTC-REQ-098	Any plateau diagnostic shall declare coordinate, segment, estimator, support, masks, uncertainty, validity, and whether it is inert, advisory, or a selected policy input.

Area	ID	Normative statement
Shift	SCI-RTC-REQ-099	Multiple level shifts shall retain ordered distinct event identities, including overlapping guards and short intervening segments, and shall follow a selected conflict and minimum-segment policy.
Leakage	SCI-RTC-REQ-100	When filtering or atmospheric-template removal can change an x/r leakage estimate, pre- and post-operation diagnostics shall be separately identified and the comparison domain and response shall be declared.
Plateau	SCI-RTC-REQ-101	Plateau or level-shift estimates shall not authorize cross-segment offset stitching in v0.1; such stitching requires separate operator, response, covariance, provenance, and validation authority.
State	SCI-RTC-REQ-102	Pair admission, mapping, leakage estimation, level-shift detection, transition masking, segmentation, reset, plateau evaluation, and optional atmospheric-template removal shall expose requested, effective, resolved, realized, unavailable, and failed states with typed causes as applicable.
CAL	SCI-RTC-REQ-103	SCI-CAL shall receive only the exact conditioned x member plus the complete paired RTC bundle needed for traceability; the r member shall not receive SCI-CAL calibration or target-atmosphere correction.
Atm.	SCI-RTC-REQ-104	The owner-selected RTC atmospheric-template operation shall precede temporal filtering; its noncommutation with every later enabled filter shall be represented by an exact composed response on the admitted domain or by an owner-resolved quantitative bound and typed unavailable/failure state.
Role	SCI-RTC-REQ-105	Every role plan shall disposition paired mapping, x/r validity, leakage diagnostics, level shifts, segmentation/reset, and atmosphere treatment; optional operation classes shall remain disabled unless explicitly selected. Science consumption is “SCI-CAL handoff and any explicitly authorized raw-domain consumers; subsequent PTC, VAL, MAP, and FLT use follows their own contracts.”

F Falsifiable predictions and limiting cases

Each prediction is an algebraic or semantic falsification target. A future conformance activity may choose an exact numerical tolerance justified by the operation and representation before observing candidate output. No result is reported by this document.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-001	Identity chain: paired raw role, $A = I$, no replacement or filter, $M = 1$, valid original pairs. Both values, raw coordinate identity, pair order, cardinality, response, flags, and support are unchanged.	Missing/reordered partner, hidden calibration, value/order/cardinality change, flag loss, nonidentity kernel, or new claim of timing/coordinate authority.
SCI-RTC-PRED-002	Role separation: feed the same paired xs/rs input to Beammap, Pointing, OOF, Science, and diagnostic-only RTC roles. Both RTC outputs remain raw; a distinct calibrated x output exists only after downstream SCI-CAL.	Any role gives either coordinate calibrated or Stokes meaning, r is calibrated or substituted for x , an RTC product is labeled mJy/beam, or CAL output is silently interchanged with RTC output.
SCI-RTC-PRED-003	Raw donor direction: choose $\phi_q = 2$, $\phi_d = 4$, and donor $x_q = 3$. Equation (SCI-RTC-EQ-002) predicts raw target replacement 1.5 and equal detector-static comparison coordinates $\chi_q = \chi_d^R = 6$.	Use of ϕ_d/ϕ_q , output other than 1.5, unequal comparison coordinate, calibration claimed from χ , or dependence on legacy responsivity .
SCI-RTC-PRED-004	Raw donor availability: independently make either factor invalid, convention/domain-incompatible, occurrence-mismatched, or set $\phi_d = 0$. The flxscale route is unavailable.	Finite replacement, stale factor reuse, row-number-only binding, zero insertion, or success without separate donor authority.
SCI-RTC-PRED-005	Order: trace a paired raw input through early repair/level-shift segmentation, selected atmospheric-template removal, every temporal stage, and phase-zero sampling, then trace only x through distinct downstream SCI-CAL. The parent and response follow equation (SCI-RTC-EQ-004).	Absolute calibration or target atmosphere appears inside RTC, r enters CAL, noncommutation is omitted, flxscale donor transfer is mislabeled calibration, or the CAL parent omits the complete RTC plan.
SCI-RTC-PRED-006	Interior target/donor impulses under fixed selection. Each output column equals equation (SCI-RTC-EQ-012); donor energy appears in the declared detector column, support, scale, and phase.	Missing or circularly wrapped energy, unexplained mixing, wrong donor identity/direction, support, normalization, or phase.
SCI-RTC-PRED-007	Constant input with sufficient context. Output equals the constant times realized DC gain after the declared transient; a DC-preserving chain is exactly constant.	Drift, divide-by-zero mask normalization, edge values advertised as interior, or a false DC-preservation claim.
SCI-RTC-PRED-008	Interior step with complete context. Response equals the cumulative impulse response and settles according to DC gain and stability.	Undeclared overshoot, failure to settle, hidden state leakage, or disagreement with the impulse-derived response.
SCI-RTC-PRED-009	Affine ramp through each FIR and admitted alignment mapping. Output slope and offset follow the zeroth and first moments of the exact kernel.	Unexplained clipping, wrong phase/slope/offset, or a ramp-preservation claim without the required moments.
SCI-RTC-PRED-010	Long bin-centered and off-bin sinusoids at DC, passband interior/edge, transition, stopband, and near output Nyquist. Amplitude and phase follow equations (SCI-RTC-EQ-013) and (SCI-RTC-EQ-014).	Gain/phase outside preregistered tolerance, unrecorded frequency convention, or an aliased component labeled passband signal.
SCI-RTC-PRED-011	Sinusoids at the realized notch center and symmetric offsets under the exact rate and state. Depth, width, phase, and transient follow realized zeros/poles.	Center/depth/width mismatch, unstable pole, rate mismatch, hidden state, or an edge transient reported ordinary-valid.
SCI-RTC-PRED-012	Coordinate mask with valid inside/outside values plus invalid, unavailable, wrong-frame, out-of-domain, and ambiguously bound coordinates. Invalid classes fail the affected operation and remain distinct from outside.	Invalid becoming outside/unmasked, entering geometry, clamping to a center/edge, or loss of the distinct cause.
SCI-RTC-PRED-013	Samples just inside, exactly on, and just outside each selected mask boundary and temporal dilation. Membership follows the recorded inclusive/exclusive and dilation policy identically across execution modes.	An undocumented boundary, frame mismatch, asymmetric dilation, or sequence-dependent membership.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-014	Impulses and constants at first/last support cells and immediately inside complete FIR context. Values, guard, support, and output admission follow the selected edge rule.	Undeclared padding, truncation, wrap, renormalization, guard, or output selection.
SCI-RTC-PRED-015	Zero, impulse, and constant inputs split and re-joined at scan/chunk/state boundaries for an IIR. Results follow the declared reset/carry and terminal-state policy.	Prior-observation leakage, hidden carry/reset, unbounded non-finite poisoning, or contradiction between split and declared state semantics.
SCI-RTC-PRED-016	Lengths $0, 1, L - 1, L, L + 1$ around every support/guard threshold, with stable scan identity.	Crash, underflow, stale output, silent scan deletion/renumbering, or ordinary-valid status without required context.
SCI-RTC-PRED-017	Inject NaN, $+\infty$, and $-\infty$ separately into a sample, factor, coordinate, coefficient, and IIR state. Failure or unavailable influence follows REQ-026 exactly.	Silent zero/coercion, finite-looking output without cause, stale state recovery, or an unbounded tail reported valid.
SCI-RTC-PRED-018	Place synthesized and replaced causes first at the directly selected occurrence and then at noncenter members of full support. Direct cases are universally excluded; noncenter causes propagate unchanged and consumer eligibility follows the declared policy.	A direct case called independent, a noncenter cause erased or made universally ineligible by RTC, or consumer disposition reported without its policy.
SCI-RTC-PRED-019	For $M = 1, 2, 3, 4, 5$ and $N = 0$ through several incomplete final strides, enumerate $j = Mn$. Cardinality and selected time follow equation (SCI-RTC-EQ-011); no block mean appears.	Wrong N_y , dropped/duplicated point, nonzero phase, averaged value, timestamp misidentification, or rate mismatch.
SCI-RTC-PRED-020	Two sinusoids separated by an integer multiple of f_{out} , before and after the realized pre-filter. Folded amplitude is the sum in equation (SCI-RTC-EQ-014).	A folded term missing from response, inadequate stopband labeled alias-free, or incorrect image frequency/amplitude.
SCI-RTC-PRED-021	Reuse one donor for two targets and give neighboring outputs overlapping filter support. Equation (SCI-RTC-EQ-016b) predicts cross-target and temporal covariance when the admitted input covariance permits it.	Independent weights assigned, off-diagonal terms silently dropped, or reused donor described as additional exposure.
SCI-RTC-PRED-022	Enable a response-changing mask, donor mix, stateful filter, edge, and decimator while providing only a scalar FIR transfer. Complete response state is unavailable.	The partial scalar transfer labeled complete, or downstream use that assumes omitted stages are identity.
SCI-RTC-PRED-023	Learned candidates include at least two safe factors and one failing each astronomical-transfer, alias, sampling, realizability, and compatibility constraint. Equation (SCI-RTC-EQ-021) chooses the largest passing factor.	A smaller safe factor selected without rule, any failing factor applied, a percentile speed used for safety, or a missing numeric predicate treated as pass.
SCI-RTC-PRED-024	Serialize and restart a learned plan, then alter one of beam identity, speed support, scan set, coefficient precision, rate, phase, role, or parent. Apply rejects the mismatch.	Silent relearning, retuning, native fallback, stale-plan application, or realized state rewriting the request.
SCI-RTC-PRED-025	Execute two observations sequentially with distinct rates, constants, factors, masks, donors, states, and output parents.	Any coefficient, state, factor, donor, mask, rate, flag, or product link leaks from the first observation.
SCI-RTC-PRED-026	Toggle optional detailed provenance and inert diagnostics while holding the selected policy and realized operator fixed. Values, flags, influence, eligibility, cardinality, response, and failures are identical.	Any scientific or numerical result changes, or an observational/production claim appears from the toggle.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-027	Bind identical learned parameters first to a separate training population and then to the apply observation. The numerical fixed-state response can match, but the second plan records data dependence and parameter/selection uncertainty.	Same-observation learning represented as fixed a priori, or conditional linearity used to erase selection or parameter uncertainty.
SCI-RTC-PRED-028	Remove in turn one required learning-population identity, tolerance, uncertainty, or downstream-compatibility predicate before resolve. The dependent plan remains unresolved and apply does not begin.	Missing evidence treated as pass, a default inserted, or output represented as applied learned mode.
SCI-RTC-PRED-029	Serialize one resolved plan and attempt during apply to alter a notch center, cutoff, tap count, donor rule, or decimation factor. The mutation is rejected or belongs to a separately authorized adaptive estimator.	Hidden retuning, silent fallback, or a realized record that rewrites the resolved plan.
SCI-RTC-PRED-030	Sweep notch center error, line drift, width, depth, Q/order, initialization, direction, and overlap. Residual line and astronomical amplitude, phase, centroid, covariance, transient, and edge loss follow the exact combined response.	Only line disappearance measured, or any changed astronomical observable omitted from the plan's validity evidence.
SCI-RTC-PRED-031	Scan the same projected source at multiple speeds and directions through a fixed filter. Its temporal spectrum and retained amplitude/PSF change according to equation (SCI-RTC-EQ-023) and exact response.	One nominal cutoff or scalar beam treated as speed- and direction-independent proof of preservation.
SCI-RTC-PRED-032	Design FIR candidates over multiple tap counts and coefficient precisions. The selected N_* is the smallest candidate passing every registered transfer, alias, phase, support, guard, and realizability predicate.	Arbitrary configured taps, approximate order formula promoted to authority, or a candidate with any failed predicate selected.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-033	Inject compact and extended sources through admitted high- or band-pass candidates. Low-frequency rejection produces the predicted lost broad modes, compensated response, or negative lobes.	Unit preservation labeled flux preservation, or removed absolute/extended modes reported available without response restoration evidence.
SCI-RTC-PRED-034	Choose donor and target with compatible <code>flxscale</code> but deliberately different sky signal or beam coupling. Raw-domain convention transfer remains algebraically correct while the surrogate is not a missing-sky reconstruction.	Compatible factors claimed to prove physical donor equivalence or independent exposure.
SCI-RTC-PRED-035	Attempt Beammap donor scaling with a new <code>flxscale</code> derived from that same conditioned Beammap. The route is rejected until an identified prior factor authority is supplied.	Circular use of the new factor or absence of the prior authority in lineage.
SCI-RTC-PRED-036	Apply filters with known nonzero group delay to Pointing/Beammap scans in opposite directions. Centroid displacement follows equation (SCI-RTC-EQ-027) with opposite scan-direction sign unless time/coordinates are corrected.	Selected grid time treated as response center, wrong displacement sign, or unrecorded correction.
SCI-RTC-PRED-037	Process calibrator and target with plans that share a nominal label but differ in coefficients, rate, phase, state, edge, or decimation. Cross-plan calibration transfer is unavailable absent explicit response accounting or observational evidence.	Nominal label, preserved unit, or <code>flxscale</code> alone treated as RTC-plan compatibility.
SCI-RTC-PRED-038	Pass all deterministic coefficient and operator fixtures but withhold the required source/scan design studies. Algebraic correctness may be supported; science qualification remains unassessed.	Deterministic pass promoted to observational performance, science qualification, or production readiness.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-039	Inject one strong coherent line and no other valid line. Attempt 1 accepts and suppresses it; Attempt 2 finds no acceptable successor. The accepted-plan count remains one and Π_1 is final with a no-successor stop.	A manufactured Π_2 or evaluation product, unbounded iteration, unsupported second notch, rejected evidence lost, or no typed stop/final-plan identity.
SCI-RTC-PRED-040	Inject two unequal coherent lines so the weaker becomes identifiable only after the stronger is suppressed. A later attempt accepts one complete two-notch plan as the next Π_k and replays it once on $x^{(0)}$; output follows the predicted cumulative response.	Attempt and accepted-plan indices conflated, increment-only plan, final use of an accumulated exploratory product, missing first notch, duplicate filtering, or response disagreement.
SCI-RTC-PRED-041	Create a peak at a first-notch edge that agrees with predicted cumulative ringing or response artifact. The successor candidate is rejected and the first plan remains final.	Artifact admitted as a line, response floor ignored, or prior accepted plan invalidated solely by rejection.
SCI-RTC-PRED-042	Compare applying a one-notch plan twice with applying the intended complete successor plan once to $x^{(0)}$. Values, response, phase, edges, support, and covariance differ unless exact operator equivalence is proved.	Repeated filtering treated as replay, or equality claimed from nominal notch identity without complete operator proof.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-043	Propose a successor that fails one cumulative transfer, stability, support, or compatibility predicate. The attempt is rejected; its evidence and cause persist while the accepted-plan index and final product remain unchanged.	Rejected proposal assigned a new accepted-plan index or evaluation, failed plan applied, prior product rewritten, rejected evidence erased, or missing predicate treated as pass.
SCI-RTC-PRED-044	Alternate between proposing addition and removal of the same line or vary its parameters outside declared stability across attempts. Termination is typed oscillating or nonconvergent.	Alternation or parameter instability labeled convergence, stable plan, or successful maximum-safe refinement.
SCI-RTC-PRED-045	Supply a sequence that reaches the owner-approved maximum attempt count without satisfying plan stability. The result records maximum-attempt termination and the last accepted plan, not convergence.	Maximum count labeled stable/converged, continued iteration, attempt count confused with accepted-plan count, or disappearance of the last accepted plan.
SCI-RTC-PRED-046	Serialize and restart the final cumulative plan, then vary original parent or learning-population identity. Exact compatible restart reproduces the product; either mismatch is rejected.	Silent relearning, stale-parent application, altered plan, or acceptance of incompatible learning lineage.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-047	Supply complete paired occurrences, then remove only one r partner and separately only one x partner. Complete pairs are admitted; either incomplete occurrence fails required-input admission.	Partner synthesis, copying, zero fill, independent join, or successful admission of an incomplete pair.
SCI-RTC-PRED-048	Apply a known invertible $IQ \rightarrow xr$ mapping and serialize/restart. Both coordinates and exact pair/mapping-revision identity reproduce equation (SCI-RTC-EQ-030).	Value, order, offset, sign, detector/sample, or mapping revision changes.
SCI-RTC-PRED-049	Make x invalid and r valid, then reverse them. The pair persists and the two validity states remain independent.	One member's state copied to the other, invalid member used as valid, or pair discarded without selected policy.
SCI-RTC-PRED-050	Inject an optical signal with known nonzero r/x response. Atmosphere- and bright-source estimators recover their own admitted ζ within preregistered uncertainty.	r assumed zero, the two estimates silently merged, wrong source parent, or missing uncertainty/support.
SCI-RTC-PRED-051	Give atmosphere and bright source deliberately different leakage ratios. Two distinct products remain available for comparison without automatic reconciliation.	Averaging, substitution, or promotion to one universal leakage value without an authorized model.
SCI-RTC-PRED-052	Generate frequency-dependent leakage and fit both a scalar and admitted spectral estimator. Only the spectral product supports the frequency-dependent claim; the scalar is range-qualified.	Scalar advertised as broadband authority, missing band/support, or absent response status.
SCI-RTC-PRED-053	Attempt to correct x from r , calibrate r , use r as an x donor, or replace invalid x with valid r . Every route is unavailable in v0.1.	Any route executes or is implied by shared pair identity.
SCI-RTC-PRED-054	Simulate correlated x/r noise and a shared selector. Joint covariance retains the cross block and selection term.	Block-diagonal independence, zero cross term, or separate selection represented as joint truth without evidence.
SCI-RTC-PRED-055	Add one paired additive level shift with known transition and unequal x/r amplitudes after isolated spikes. Detection follows the repaired-stream model and reports the event, both amplitudes, transition mask, guards, and two segments.	Shift absorbed as spikes, amplitude copied between coordinates, missing mask/segment, or correction claimed from detection alone.
SCI-RTC-PRED-056	Place isolated spikes immediately before and after a level shift. The selected despiked-then-shift suborder yields the preregistered event and repaired values.	Undeclared suborder, shift estimator using unrepaired spikes, or circular repair with no recorded plan.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-057	Put a filter impulse on each side of a declared level-shift boundary. Responses reset or carry exactly as selected and never silently cross segments.	Cross-boundary state leakage, ordinary-interior labels in guards, or missing reset provenance.
SCI-RTC-PRED-058	Insert two level shifts with overlapping guards and a short middle segment. Event identities remain ordered and the selected conflict/minimum-segment policy gives one deterministic disposition.	Event collapse, order dependence outside policy, negative/ambiguous segment, or silent guard truncation.
SCI-RTC-PRED-059	Evaluate plateau diagnostics on known pre/post levels, then toggle their advisory output while holding policy fixed. Values and segmentation are unchanged and no stitching occurs.	Plateau toggle changes science values, silently stitches offsets, or erases the level-shift cause.
SCI-RTC-PRED-060	Measure leakage before and after a filter whose x and r responses differ. The two diagnostics differ according to the composed response and retain separate identities.	One value reused across stages, filtering treated as commuting without proof, or pre/post parentage lost.
SCI-RTC-PRED-061	Apply the selected RTC atmospheric template before a noncommuting temporal filter and compare with the reversed order. The authorized order matches its exact composed response or preregistered bound.	Reversed result treated as equal, missing response/bound, or target SCI-CAL atmosphere conflated with the RTC template.
SCI-RTC-PRED-062	Stop iterative refinement before $A_{\max}$ after A actual attempts. The record satisfies $K \leq A < A_{\max}$ and contains no fabricated attempts.	$A = A_{\max}$ forced, artificial no-op attempts, or accepted-plan count confused with either attempt quantity.
SCI-RTC-PRED-063	For every Beammap, Pointing, OOF, Science, and diagnostic-only role, inspect the resolved plan and output bundle. Paired mapping, validity, leakage, level-shift/segment/reset, atmosphere treatment, and consumer routing are explicit; optional classes remain disabled unless selected.	Missing disposition, role-dependent loss of the pair, optional operation enabled by implication, or Science routing that bypasses its own downstream contracts.

F.1 Analytic limits

The following limits are required: a compatible donor pair with $\phi_q = \phi_d$ gives unit raw donor scale; a replacement indicator tending to zero gives identity; filter coefficients tending to a Kronecker delta give identity; $M \rightarrow 1$ gives no rate change; a source-mask radius tending to zero selects only exact valid coincidences under the declared boundary; complete context approaches the declared interior response; and a causal pole approaching the unit circle exposes increasing memory and never retains a stability claim at or outside the circle.

End of shared normative core: SCI-RTC-v0.1/r0.5. View-specific material cannot add to or modify this scientific authority.